Hydrological Impact analysis of Sir Arthur Cotton Barrage

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Civil Engineering

By

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Under the guidance

of

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CERTIFICATE

This is to certify that the work contained in this project titled **"Hydrological Impact analysis of Sir Arthur Cotton Barrage"** is a bona fide work carried out by **Ms. Sunkara Lakshmi Srija**, in the Department of Civil and Environmental Engineering, Indian Institute of Technology Patna, under my supervision and that it has not been submitted elsewhere for the award of any degree.

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DECLARATION

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Signature of the student:

Project title: Hydrological Impact Analysis of Sir Arthur Cotton Barrage

This is to certify that Ms. Sunkara Lakshmi Srija.

1. has sincerely worked on this project
2. has contacted me regularly to update on the progress of this project
3. has received my comments on the preliminary version of the report and permission and will address those before the final submission /presentation
4. may be allowed to present/defend the project before the department

Signature of the Project Guide:

Name of the Project Guide: Dr. Om Prakash

Date:

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I would like to extend my heartfelt thanks to Dr. Om Prakash, Associate Professor, Department of Civil and Environmental Engineering, Indian Institute of Technology Patna, who guided and supported me during the execution of this project. He is an expert in the field of hydrology and water resources engineering, and he showed me the way to accomplish the task at hand.

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ABSTRACT

In recent decades, hydrological processes within river basins all across the globe have experienced remarkable changes, which are mostly the result of two main factors, the first being changes in climate and the other being increased anthropogenic influence on nature. Changes in rainfall regimes, land use/cover changes, and the fast growth of urbanized areas have all contributed to a fundamental change in how water travels within the network of rivers. The transition of rainfall into runoff is one of the most basic principles of the hydrological cycle; however, it is not as simple as it seems. Hydrological processes are subject to constant changes caused by the dynamics of rainfall intensity and amount, state of soil, and presence of moisture in it. The interaction of these factors creates certain unpredictability, and thus, it becomes increasingly difficult to predict rainfall-runoff relationships.

Given the extensive nature of the Godavari River basin in India, which covers an estimated catchment area of about 302,065 km², such variations can be analyzed using this area. This area contains diverse geographical regions ranging from high plateau regions to flatlands found at the lower reaches of the river, and the river's hydrology relies greatly on the southwest monsoon season. It means that any significant change in precipitation and land surface characteristics in this region must leave traces in the flow regime of rivers in its lower reaches. In this regard, the Sir Arthur Cotton (SAC) barrage located at Dowleswaram in Andhra Pradesh occupies the lower reaches of this extensive river system. The water levels in this barrage represent the cumulative effect of all processes occurring within this river basin. Thus, any change in precipitation and land use patterns in this river basin can impact the flow regime in the SAC barrage.

This has been the driving force behind the current research that seeks to conduct an intensive study of the Godavari catchment in order to evaluate how alterations in land use and the monsoon rainfall regime have been affecting the process of runoff formation and streamflow dynamics at the SAC Barrage. Geospatial tools in ArcGIS are used to map the physical characteristics of the basin, including its terrain, drainage patterns, soil types, and land cover, while also tracking how land use has changed between 2017 and 2024. The SCS Curve Number method is then applied to translate these land cover changes into their hydrological consequences, specifically in terms of how much more or less runoff the basin is likely to produce. Rainfall trends across 49 stations over a 37-year period from 1988 to 2024 are examined using Sen's Slope estimator to detect whether the monsoon has been getting stronger, weaker, or more variable over time.

For rainfall-inflow prediction, two data-driven modeling approaches are developed and comparatively evaluated, namely Long Short-Term Memory (LSTM) deep learning networks, and Genetic Programming (GP). These models incorporate a 5-day lag structure and engineered hydrological features including antecedent rainfall indices, rate of change of inflow, and cumulative rainfall variables to capture the temporal dynamics of the rainfall–runoff process. Sensitivity analysis is carried out to establish the most hydrologically significant parameters that influence inflow variations at the SAC Barrage.

In summary, the entire study is geared toward giving an understanding of the hydrologic processes that control the Godavari Basin, which will not only help in the estimation of inflows at the SAC Barrage but also aid in comprehending the responses of this extensive and complicated basin system to current environmental changes.

Keywords: Godavari Basin, Sir Arthur Cotton Barrage, Rainfall-Runoff Modeling, LULC Change, SCS Curve Number, Sen's Slope, LSTM, Genetic Programming, Monsoon Hydrology, Non-Stationarity, ArcGIS

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Acronyms

ArcGIS - Geographic Information System software by Esri

CWC - Central Water Commission

CN - Curve Number

DEM - Digital Elevation Model

GP - Genetic Programming

HSG - Hydrologic Soil Group

India-WRIS - India Water Resources Information System

LULC - Land Use / Land Cover

LSTM - Long Short-Term Memory

MAE - Mean Absolute Error

MAPE - Mean Absolute Percentage Error

NSE - Nash–Sutcliffe Efficiency

R^2 - Coefficient of Determination

RMSE - Root Mean Square Error

SAC Barrage - Sir Arthur Cotton Barrage

SCS-CN - Soil Conservation Service Curve Number

Sen's Slope - Non-parametric trend magnitude estimator

Chapter 1

INTRODUCTION

1.1. Overview

Water resources systems across the world are undergoing significant stress due to the combined influence of climate variability and human-induced changes. In recent decades, alterations in rainfall patterns, land use/land cover (LULC), and rapid urbanization have led to noticeable changes in hydrological processes, particularly in runoff generation and streamflow regimes. Research has shown that almost 24% of the world's rivers have undergone major modifications in terms of their flow regime, influenced by climate variables like changes in precipitation patterns and increases in temperature, as well as anthropogenic activities like damming, urbanization, deforestation, and water withdrawal.

One of the key problems in contemporary hydrology is the increasing non-stationarity of hydrologic systems, in which rainfall-runoff relations cannot be considered stationary anymore in face of variations in land surface and climatic dynamics. Rainfall-runoff processes are highly complex and non-linear, with their functioning determined by an ever-changing set of variables comprising rainfall characteristics, soil moisture levels, vegetation cover, and pre-event catchment saturation.

1.2. Global Hydrological Concern

1.2.1. Climate Variability and Rainfall Changes

The most pressing issue that exists in today's world is the modification of precipitation regimes, in terms of their frequency, intensity, and pattern. Irregular rainfall has been witnessed by many locations, along with extreme storms and droughts, which affect the process of runoff formation and discharge into rivers. It results in higher flood occurrences and lower base flows, which increases uncertainties regarding the use of irrigation waters.

1.2.2. Land Use / Land Cover (LULC) Changes

Urbanization, deforestation, and agricultural development have altered the natural properties of the catchment. Figure 1.1 demonstrates how the forested area changed between 2000 and 2020 due to widespread deforestation in the tropics and subtropics. (Potapov et al., 2022). Concurrently, Figure 1.2 illustrates the expansion of croplands between 2000 and 2019, highlighting the direct conversion of forested land to agriculture as a dominant driver of land cover transformation. These land cover transitions increase peak discharge, reduce groundwater recharge, and alter natural drainage patterns - sometimes exceeding the influence of climatic factors on runoff generation.

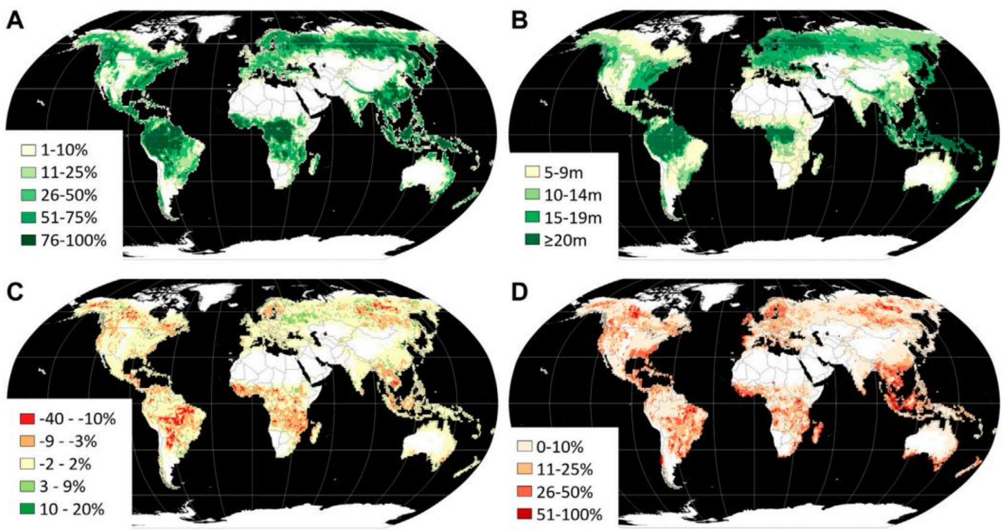


Figure 1.1. Forest extent, structure, and dynamics for each 1° × 1° grid cell (A–D).
(A) Forest area in 2020 (% cell area), (B) Mean forest height in 2020 (m),
(C) Net forest area change (2000–2020), (% cell area),
(D) Forest loss, disturbance, and degradation (2000–2020), (% of year 2000 forest area).
Source: Potapov et al. (2022)

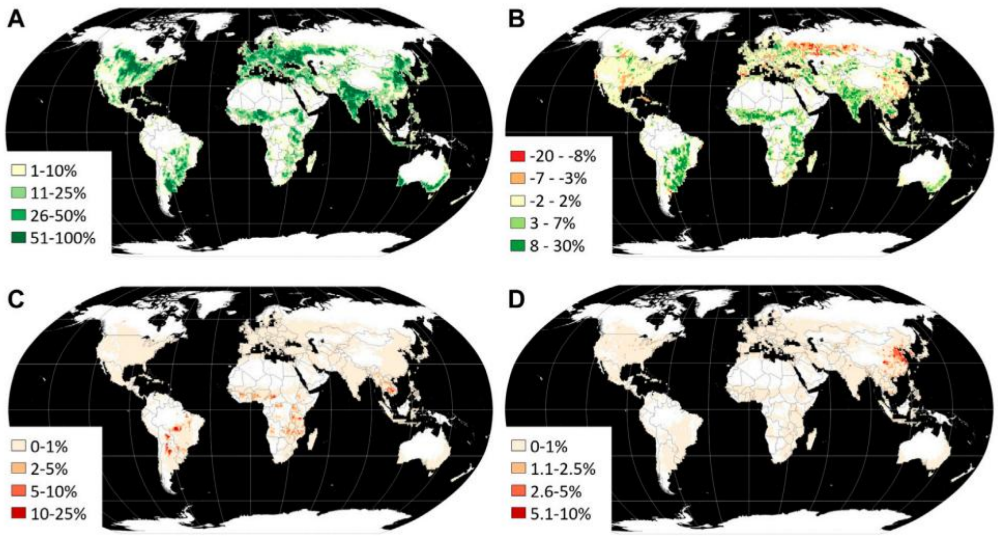


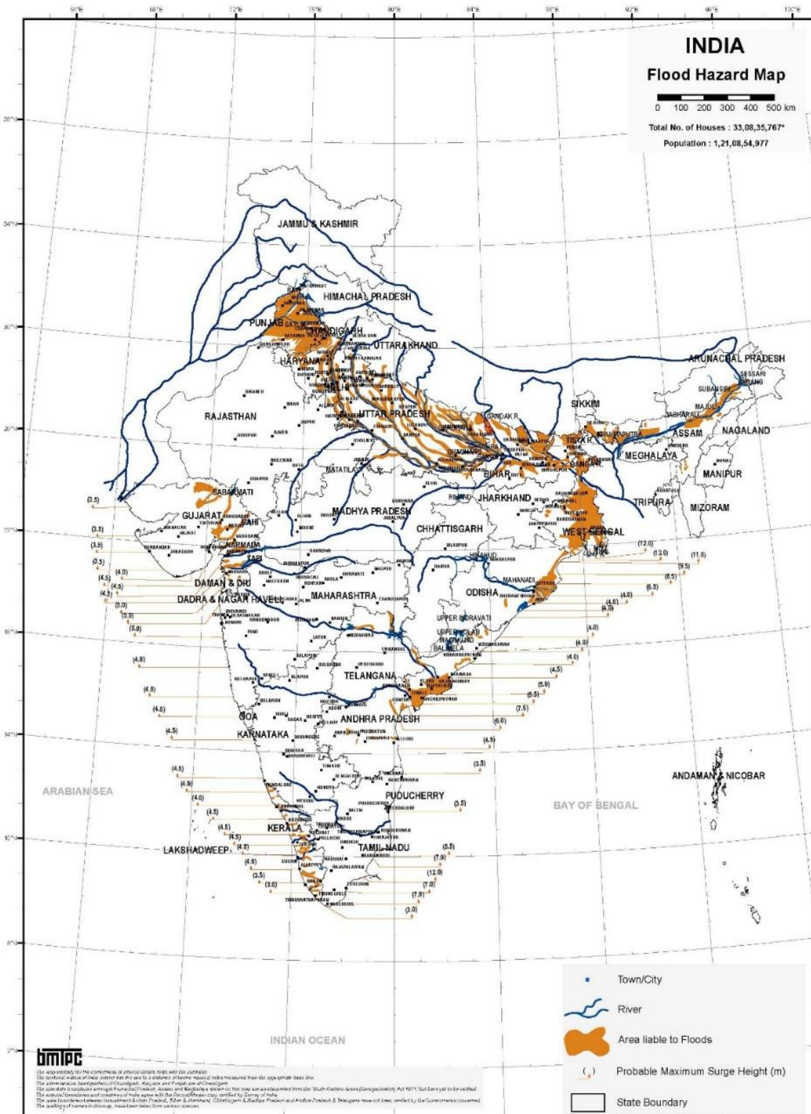
Figure 1.2. Cropland extent and dynamics for each 1° × 1° grid cell (A–D).
(A) Cropland area in 2019 (% cell area)
(B) Net cropland area change (2000–2019), (% cell area)
(C) Cropland gain (2000–2019) within forest loss areas (2000–2020),(% cropland gain area within a cell),
(D) Cropland loss (2000–2019) within 2020 built-up lands (% cropland loss area within a cell).
Source: Potapov et al. (2022)

1.2.3. Non-Stationarity of Hydrological Systems

Traditionally, hydrological analysis assumed stationarity, meaning past data could reliably predict future behavior. However, due to climate change and land surface alterations, this assumption is no longer valid. The rainfall–runoff relationship has become non-linear and temporally non-stationary, reducing the reliability of conventional hydrological models and increasing uncertainty in flood forecasting and long-term water resource planning.

1.3. Hydrological Scenario in India

The hydrology in India is predominantly influenced by monsoon, which accounts for 75-80 percent of annual precipitation during the brief period from June to September, resulting in significant variations in the timing and distribution of water supply. In the past few decades, there have been evident changes in the patterns of monsoons in India in the form of unpredictable monsoon onset, increased intensity of rainfall, prolonged dry spells, and frequent floods. At the same time, there has been an increase in urbanization, intensive agriculture, and deforestation, which have impacted the characteristics of soil, increased surface runoff, and decreased infiltration capacity. The Indian river systems are controlled by dams, barrages, and canals, and though this has increased water efficiency, it has also changed the natural pattern of flows and made hydrological forecasting difficult. All these factors make the need for hydrological modeling indispensable. (Figure 1.3).



BMTPC : Vulnerability Atlas - 3rd Edition; Peer Group, MoHUA; Map is Based on digitised data of SOI, GOI; Census of India 2011; Flood Atlas (1987),Task Force Report (2004), C.W.C., G.O.I. Houses/Population as per Census 2011; * Houses including vacant & locked houses. Disclaimer: The maps are solely for thematic presentation.

Figure 1.3. Map of Flood Hazards in India indicating the flood hazard zones and maximum possible storm surge heights.

Reference: BMTPC (2016). Vulnerability Atlas of India, 3rd Edition. Ministry of Housing and Urban Affairs (MoHUA), Government of India.

1.4. Study Area Selection - Godavari Basin and SAC Barrage

Godavari Basin was chosen because of its size, hydrological complexity, and abundant network of rain gauges. The SAC Barrage at Dowleswaram can serve as the best location to discharge water based on the combined effect from the entire basin. The study site has been described thoroughly in Chapter 3.

1.5. Objectives of the Study

The main objective of this study is to analyze the hydrological effect of rainfall variations, change in land use and land cover (LULC) and soil properties on the streamflow at Sir Arthur Cotton Barrage. More specifically, the objectives include:

1. **Rainfall Trend Analysis**

   Analysis of spatio-temporal variability of the monsoon rainfall in the Godavari Basin for several decades (1988 - 2024). Analysis of long term trends and detection of extreme rainfall events using non-parametric statistics.

2. **Catchment Characteristic Analysis**

   Mapping and analysis of changes in LULC from 2017 - 2024 and their hydrological implications. Identification of hydrologic soil group based on soils in the catchment and analysis of their effect along with LULC on runoff formation.

3. **Surface Runoff Analysis**

   Analysis of surface runoff generation in relation to land cover change and rainfall variations using SCS-CN method.

4. **Data-Driven Inflow Modeling**

   Development and testing of various machine learning techniques for daily inflow prediction at SAC Barrage using spatially distributed monsoon rainfall data.
   Evaluation of the performance of different modeling approaches on independent data using conventional hydrological performance measures.

5. **Comparative Analysis**

   Comparing the forecasting capability and explain ability of the proposed models for simulating rainfall-inflow relations at SAC Barrage

6. **Sustainability Assessment**

   For an evaluation of the possible consequences of the observed hydrological changes concerning sustainable water management for the Sir Arthur Cotton Barrage.

   For making recommendations regarding basin and barrage management in view of the results of this study.

1.6. Organization of the Thesis

The thesis comprises of seven chapters and is described as follows:

- **Chapter 1: Introduction** discusses the background of the study, problem statement, motivation, scope and objectives.
- **Chapter 2: Literature Review** analyses existing studies about rainfall-runoff modelling, LSTM Deep Learning techniques, Genetic Programming and hydrological studies in relation to Godavari basin.
- **Chapter 3: Study Area Description** provides comprehensive description of Godavari basin and its geographical, climatic and hydrological features, as well as Sir Arthur Cotton Barrage.
- **Chapter 4: Data Collection & Methodology** explains the data employed such as rainfall and inflow data and the methodology applied that includes use of SCS-CN method, LSTM and Genetic Programming algorithms.
- **Chapter 5: Results and Discussion** discusses rainfall trend analysis, runoff estimates along with results and discussion of LSTM and GP models based on statistical and graphical interpretation.
- **Chapter 6: Comparative Analysis** compares and analyzes the prediction performances of LSTM and Genetic Programming models for inflow prediction at the Barrage.
- **Chapter 7: Conclusions and Recommendations** discusses conclusions drawn based on study results and suggestions for further work and sustainable water management practices.

Chapter 2

2. LITERATURE REVIEW

2.1. Background

The ability to accurately forecast the streamflow and runoff processes is critical for planning in water resources, flood forecasting, and management in river basins. In this regard, many techniques have evolved throughout the years in an attempt to comprehend and model the process of rainfall-runoff. The selection of a suitable technique depends largely on the amount and type of available data, the size of the catchment, and the purpose of the study. For large river basins such as the Godavari, where rainfall varies significantly both temporally and spatially, and the response of the basin is nonlinear, the conventional methods alone may not be sufficient. This chapter discusses the modeling techniques that are pertinent to the current research, which include runoff estimation techniques, the SCS Curve Number method, artificial neural networks, Long Short-Term Memory deep learning models, and Genetic Programming.

2.2. Runoff Estimation Methods

There are several techniques used to estimate surface runoff generated by precipitation which can be categorized into four broad types:

1. Empirical Techniques

- **Rational Method (Q = CiA):** Peak flow rate estimation by using rainfall depth, catchment area, and runoff coefficient. Suitable for use only in small urban watersheds less than 200-500 acres in area with uniform rainfall.
- **Regional Empirical Formula**: Including Khosla's formula, Inglis and DeSouza formula, Strange's table, and Binnie's percentages for specific geographical regions of India. These techniques are useful for giving rapid estimates that can't be generalized beyond that region.

- **Cook's Method**: Peak surface runoff estimation using four catchment attributes (relief, soil infiltration capacity, cover, and surface retention). This method does not allow quantitative estimation of catchment attributes.
- **US Soil Conservation Service (SCS) Curve Number (CN) Method**: Direct runoff estimation from storm rainfall depending upon the land use, soil type, and antecedent moisture condition of the catchment (Subramanya, 2008; Verma et al., 2021).

2.Hydrological Modeling and Hydrograph Methods

- The **Unit Hydrograph Approach** models the pattern of runoff flow by assuming linearity in the watershed's response to a unit amount of effective rainfall. The approach produces a time series of runoff and needs good parameter calibration.
- The **Infiltration Approach** includes approaches like the Φ-Index and W-Index which derive runoff by using total rainfall minus infiltration capacity.
- The **Overland Flow Hydrograph** Approach studies the behavior of water flowing on the land prior to channelization.

3. Direct Measurement Methods

- **Velocity-Area Approach** estimates discharge by determining the speed of flow and the cross-sectional area of the channel. Limited to gauged rivers due to the presence of gauging stations.
- Weirs and Notches are physical devices used to determine the flow rate by introducing these devices into the channel. Ideal for experimental catchments but cannot be applied to river basins.

4. Advanced and Geospatial Methods

- The GIS and Remote Sensing Approach utilizes a GIS-based technique where maps of soils, land use and topographic features are superimposed to spatially allocate runoff estimation in the basin for basin-wide analysis at a fine level of resolution (Verma et al., 2021).
- The Hydrological Modeling Approach uses software like HEC-HMS and SWAT to simulate the rainfall-runoff processes in a model form, which could be lumped, semi-distributed or distributed models that provide physically based representations of the hydrological system and need a lot of calibration data.

- The Data Driven Approach is based on algorithms like ANN, LSTM network and GP that learn the rainfall-runoff process relationship from historical data sets without explicitly using the physics behind it (Antar et al., 2006; Liong et al., 2002; Yokoo et al., [2021]).

2.2.1. SCS Curve Number Method

The SCS Curve Number approach continues to be among the most commonly applied empirical methods for determining direct runoff generated from rainfall events. The method combines land surface characteristics, soil characteristics, and antecedent moisture levels into a single value called the Curve Number. As Verma et al. (2021) point out, although the method provides satisfactory results for moderate rainfall events, it underestimates runoff during large rainfall events. In addition, according to Bhadra et al. (2009), the method is unable to account for time, making its application in simulating continuous daily flows difficult. Despite these limitations, the SCS-CN approach was employed in the current study to determine the physical relationship between rainfall and runoff in the Godavari Basin.

2.2.2. Machine Learning Approaches in Rainfall-Runoff Modeling

The **Artificial Neural Network** is one of the first learning models used in rainfall-runoff models. Antar et al. (2006) found out that the feedforward neural network can successfully simulate streamflow from the Blue Nile catchment, providing better results than other regression methods since the model can handle nonlinearity between input and output. According to Gholami and Sahour (2021), ANNs can generalize despite the lack of sufficient data. Additionally, Nourani (2017) suggested using the Emotional ANN, which proved its advantages in extreme conditions. Yet, classical ANNs process each time step individually.

The **Long Short-Term Memory** networks, designed by Hochreiter and Schmidhuber (1997), overcome this issue through their memory retention over several time periods due to a gating process. Yokoo et al. (2021) showed that the LSTM networks were capable of implicitly learning the hydrological conditions, including soil moisture and groundwater storage, better than conceptual models for basins showing nonlinear behavior. Their capacity to consider spatially distributed precipitation data and simulate extreme peak discharges make them ideal for inflow forecasting in large barrages.

Genetic Programming (GP) was proposed by Koza (1992), which is a type of evolutionary computation approach that automatically generates explicit mathematical equations for modeling the relationship between rainfall and runoff. Liong et al. (2002) found that the equations generated using GP were physically consistent and exhibited prediction accuracy equivalent to neural networks. Further, Savic et al. (1999) showed that the equations developed using GP had the same level of accuracy as ANN but were also comprehensible.

2.3. Land Use/Land Cover Change and Hydrological Implications

Alteration in land use and land cover has been identified as one of the key causes of hydrological modifications within a river basin. According to Potapov et al., there were many changes in land cover worldwide from the year 2000 to 2020, which include deforestation and expansion in agriculture. This led to more runoff on the surface and less infiltration rate. Yang et al. pointed out that the modification in runoff generation due to land use land cover (LULC) change together with variation in precipitation was of immense importance within major river basins.

2.4. Research Gaps and Motivation for the Present Study

Literature review reveals the following main research gaps that inspire the current study:

1. Most rainfall-runoff investigations conducted in Indian river basins are dependent on lumped conceptual or linear regression models that cannot account for spatial rainfall differences and non-linear inflow dynamics (Bhadra et al., 2009; Subramanya, 2008).
2. The use of LSTM models for forecasting inflow in Indian barrages through spatial rainfall and antecedent precipitation measurements has not been adequately addressed, although they have been found effective in other parts of the world (Yokoo et al., 2021).
3. Comparison between GP and LSTM techniques for rainfall-runoff analysis has not been carried out sufficiently in India, although GP has been considered promising internationally (Liong et al., 2002; Savic et al., 1999).

4. There is no comprehensive framework that considers the impact of LULC alteration, soil characteristics, and rainfall variations on the generation of runoff in the Sir Arthur Cotton Barrage (Potapov et al., 2022; Verma et al., 2021).

In order to fill these knowledge gaps, the current research work introduces a hybrid modeling approach that incorporates the physical principles of the SCS-CN method (USDA, 1972), the predictive capabilities of LSTM (Hochreiter and Schmidhuber, 1997), and the interpretability of GP (Koza, 1992) using hydro-meteorological data collected over four decades in the Godavari Basin.

2.5. Summary

The main concepts regarding rainfall-runoff models applicable in the current research were discussed in this chapter. Firstly, SCS-CN is a physically-based model but is temporal in nature. Secondly, the machine learning approach, especially LSTM and GP, has advantages in terms of accuracy and interpretability. Thirdly, literature on land use and land cover changes indicated that the impact of land surface changes has been increasing. On this basis, the current research will employ a mixed approach for daily inflow forecasting in the Sir Arthur Cotton Barrage.

Chapter 3

3. STUDY AREA

3.1. Overview

The present study focuses on the Godavari River basin in peninsular India, with the Sir Arthur Cotton (SAC) Barrage at Dowleswaram, Andhra Pradesh serving as the primary hydrological outlet and point of analysis. The basin represents one of the most hydrologically significant and complex river systems in India, encompassing diverse topographic, climatic, and land surface conditions that collectively govern the rainfall-runoff dynamics investigated in this study. This chapter describes the geographical, climatic, soil, and infrastructural characteristics of the study area to establish the physical context for the hydrological analysis and modeling undertaken in subsequent chapters.

3.2. Godavari Basin - Location and Extent

The Godavari River Basin forms the second-largest river basin in India, spanning an area of around 302,065 km^2, which forms about 9.5% of the total geographical area of the country. It spans seven Indian states including Maharashtra, Andhra Pradesh, Chhattisgarh, Odisha, Madhya Pradesh, Karnataka, and parts of the Union Territory of Puducherry. It forms one of the most important river basins of peninsular India (CWC & NRSC, 2014).

The Godavari River arises from the Western Ghats in the district of Nashik, Maharashtra, near Trimbakeshwar, and drains into the Bay of Bengal through a large delta near the city of Rajahmundry in the state of Andhra Pradesh after traversing a distance of around 1,465 km eastwards through the Deccan plateau. There are eight major sub-basins and 466 watersheds in the Godavari River Basin.

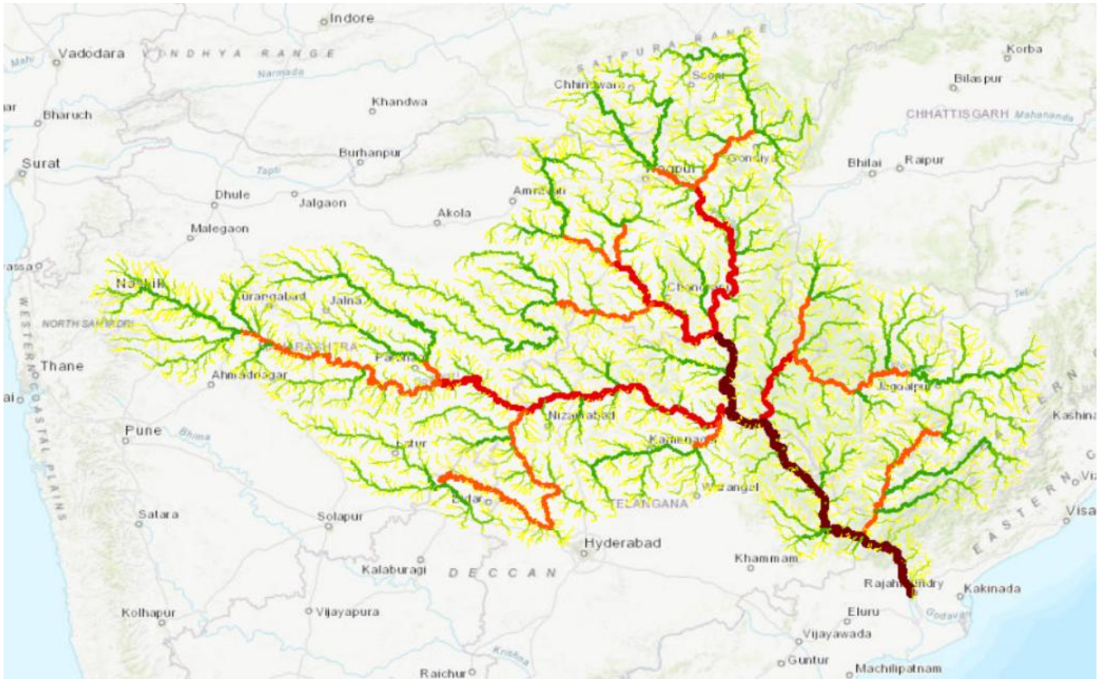


Figure 3.1: Location map of the Godavari Basin showing river network

Source: ArcGIS

3.3. Topography and Drainage

The topography in the Godavari basin differs significantly from west to east. In the upstream area of the basin, the topography comprises the hilly regions of the Western Ghats and the Deccan Plateau, with elevations higher than 1,000 meters above mean sea level. In the middle area of the basin, the landscape includes slightly rolling regions of plateaus, while the lower part of the basin merges into alluvial plains that form the Godavari Delta, which eventually drains into the Bay of Bengal.

The drainage system is well-established with several important tributaries such as the Pravara, Manjra, Wardha, Wainganga, Pranhita, Indravati, and Sabari rivers merging with the main river channel from the northern and southern parts of the basin. The confluence of these rivers in the mid and lower regions of the basin causes an increase in the volume of floodwater that flows downstream to the SAC Barrage, rendering the inflow estimation process at this point very significant.

3.4. Climate and Rainfall

The climatic conditions in the Godavari River basin can be described as mainly tropical, influenced by the southwest monsoon. The southwest monsoon occurs between the months of June and October, providing over 80 percent of the total annual rainfall in the basin. Due to the high concentration of rainfall in time, there is a lot of seasonal variation in the amount of water flowing in the rivers.

The average annual rainfall in the basin is around 1,096 millimeters, with high variations in space. The rainfall varies from about 700 millimeters in the semiarid west parts of the basin to over 1,500 millimeters in the humid east parts of the basin. The spatial variation in the rainfall pattern explains the differences in the runoff patterns within the basin. For this reason, a good inflow model should account for the spatial variation in rainfall.

3.5. Rain Gauge Network

Rainfall data used in this research was extracted from 49 rain gauge stations located in the Godavari basin between 1988 and 2024. The selection of stations aimed at providing sufficient spatial representation of rainfall over the basin covering an area of 302,065 km. The distribution of such stations is illustrated in Figure 3.2.

This level of station distribution ensures a representative input of basin rainfall data, which is a major driving force for the two models (LSTM and GP models) under investigation.

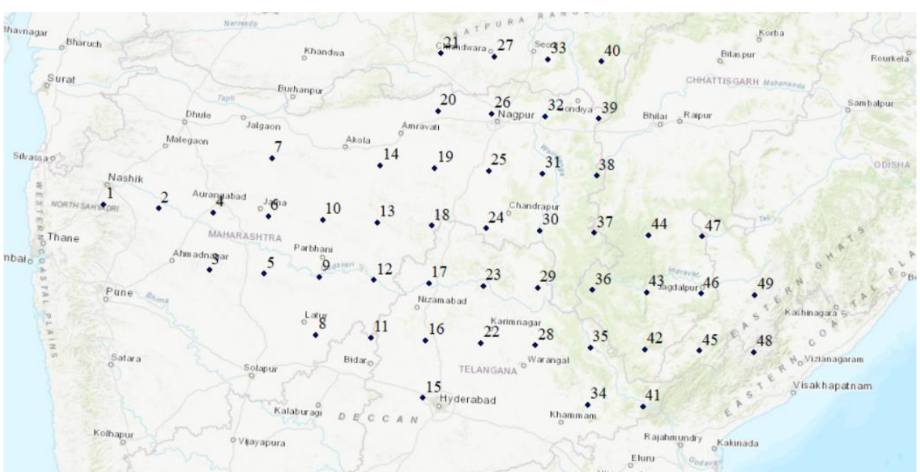


Figure 3.2: Spatial distribution of 49 rain gauge stations across the Godavari Basin

3.6. Sir Arthur Cotton Barrage - Location and Significance

Sir Arthur Cotton (SAC) Barrage is situated at Dowleswaram near Rajahmundry in East Godavari District of Andhra Pradesh at latitude 17.00°N and longitude 81.80°E. It lies at the most downstream location of the Godavari River as it flows through the Deccan Plateau prior to spreading into its distributaries in the deltaic region before draining into the Bay of Bengal.

Figure 3.3: Location map of the Sir Arthur Cotton Barrage at Dowleswaram showing upstream contributing catchment area and major tributaries

Historical Context

Construction of the barrage took place in 1852 under the direction of Colonel Sir Arthur Thomas Cotton of Madras Engineers, thus making it one of the oldest existing barrages in India. Its construction was based on the need to divert flows from the Godavari River into the canal system in the Krishna-Godavari delta to irrigate the arable land that had been experiencing drought and famines. Successful implementation of the project resulted in the transformation of agriculture in the area and the establishment of an irrigation system that exists even today in Andhra Pradesh.

Physical Attributes

The barrage has a length of about 1,100 metres stretching across the Godavari River and can accommodate flood flows that exceed 3.5 million cusecs, making it one of the largest barrages in terms of capacity.

Operational Importance

The SAC Barrage acts as a control mechanism that regulates the distribution of water into both the Western and Eastern Delta canal networks of the Godavari River system. This helps to irrigate approximately 10 lakh hectares of agricultural land in the East Godavari, West Godavari, and Krishna districts. Moreover, the SAC Barrage is used as a flood control barrier of the Godavari and managing operations at this barraging structure during the peak flow discharge during the monsoon period is crucial to avoid flooding of the heavily populated delta regions.

Hydrological Significance for This Study

In terms of hydrology, the SAC Barrage holds a unique importance by being an integrating node of the entire upstream basin. Any variation in rainfall pattern, land cover type, soil moisture status, or upstream water storage occurring anywhere within the 302,065 km2 Godavari basin results in a corresponding variation in flows measured at this barrage. The SAC Barrage becomes a very suitable location, from a scientific point of view, for conducting the present research.

3.7. Summary

The attributes related to the geographical, topographical, climatological, and infrastructure conditions of the Godavari basin and SAC Barrage were explained in this chapter. The combination of a large size, variable topography, predominantly monsoon conditions, and the critical location of the SAC Barrage downstream made it an ideal choice as the study region for the current research study. Data from 49 rain gauges form the rainfall dataset used as the main source of data for the modeling process in Chapters 4 and 5.

Chapter 4

4. DATA COLLECTION AND METHODOLGY

4.1. Data sources

These datasets have been used for the purpose of this analysis:

- Daily rainfall record data (1988-2024) recorded from 49 rainfall stations within the Godavari basin - Indian Metrological department.
- Daily Grand Total Inflow data at the Sir Arthur Cotton Barrage (1988-2024) provided by Barrage Authority.
- LULC maps for 2017 and 2024 derived from Bhuvan Portal and processed in ArcGIS.
- SRTM 30-m DEM for basin delineation downloaded from the USGS Earth Explorer portal.
- Soil classification data obtained from National Bureau of Soil Survey.

4.2. Rainfall Trend Analysis Using Sen's Slope Estimator

For determining the trend and quantifying its magnitude with respect to rainfall and basin runoff potential, Sen's Slope Estimator (Sen, 1968) was used along with the MK test (Mann, 1945; Kendall, 1975). These methodological approaches are considered the standard techniques suggested by the World Meteorological Organization (WMO) and Central Water Commission (CWC) for hydro-meteorological trend analysis, and is also commonly used in Indian hydrological studies (Jain & Kumar, 2012; Kumar et al., 2009).

Mann-Kendall Trend Test: The MK test determines if there is a significant monotonic trend in the time series data without assuming normal distribution. The statistic S is obtained by comparing the values of each pair, while the standardized statistic Z is calculated using Var(S) after considering any tied observations. Trend significance was tested at $\alpha = 0.05$ (95% confidence); $|Z| > Z_{(\alpha/2)}$ indicates the rejection of the null hypothesis of no trend.

Sen's Slope Estimate: The magnitude of the trend was determined as the median value of all slopes between data pairs:

$$Qi = \frac{xj - xk}{(j-k)} \qquad\qquad \beta = Median\,\{Qi\}$$

A positive β value represents an upward trend (mm/year), whereas a negative value represents a downward trend. The confidence interval for the trend was calculated using the normal approximation approach.

The reason why Sen's Slope was chosen over OLS regression is that Sen's Slope is highly resistant to outliers and missing data and works well with non-normally distributed data.

4.3. Land Use / Land Cover (LULC) Analysis

Land use and land cover are significant factors influencing the hydrologic behavior of a river basin. The classification of the LULC of the Godavari River Basin was performed for the years 2017 and 2024 by utilizing data available on the Bhuvan portal. It is worth mentioning that seven land cover categories were recognized within the Godavari River Basin, which include water bodies, forests, inundated vegetation, agricultural lands, urban areas, bare grounds, and range grounds.

Water Bodies – make contribution to river discharge without any infiltration losses.

Forest – decreases surface runoff because of intensive interception, infiltration and transpiration.

Flooded vegetation – plays the role of buffer for absorption of surface runoff.

Crop land – experiences seasonal variations in runoff generation from one type of crop and its development stages.

Built-up area – increases runoff greatly due to non-infiltrative surfaces of roads and buildings.

Bare ground – seals under rains making soil non-infiltrative and promoting runoff generation.

Range ground – decreases surface runoff to some extent, but not as much as forests do.

4.4. Soil Classification and Hydrological Soil Group (HSG)

The soils of the Godavari River Basin were divided into Hydrologic Soil Groups (HSGs) using the USDA soil classification system. There are four HSGs, which denote an increasing potential for runoff:

- Group A: High infiltration rate, low runoff
- Group B: Moderate infiltration, moderate runoff
- Group C: Slow infiltration, higher runoff
- Group D: Very slow infiltration, highest runoff

4.5. Selection of Runoff Estimation and Modeling Approaches

4.5.1. SCS-CN method

SCS-CN model was chosen for estimating surface runoff because of its simplicity and ease of application with small data requirements and applicability in large watersheds. This model is able to incorporate the cumulative effect of land use, soil characteristics, and antecedent moisture condition into one CN value, which makes it very suitable for the Godavari basin where GIS-based spatial information on LULC and soils is available (Subramanya, 2008; Verma et al., 2021).

SCS-CN Equations,

The basic equations of the SCS-CN method are:

$$Q = \frac{(P-0.2S)^2}{(P+0.8S)} \quad \text{for P > 0.2S, otherwise Q = 0} \tag{4.1}$$

$$S = \frac{25400}{CN} - 254 \tag{4.2}$$

where:

Q = direct runoff depth (mm)

P = total rainfall (mm)

S= Maximum possible storage after the start of runoff (mm)

CN= Curve number (dimensionless, tabled based on LULC and HSG)

The weight composite curve number was calculated for the basin as follows:

where CNi and Ai represent curve number and area respectively of sub-area i.

$$\text{CN\_composite} = \frac{\Sigma(CNi \times Ai)}{\Sigma Ai} \tag{4.3}$$

where CNi and Ai represent the curve number and area respectively of sub-area i.

4.5.2. LSTM Model

The Long Short-Term Memory (LSTM) model was selected to predict daily inflows due to its capability to capture temporal dependencies present in time-series data. Unlike conventional feedforward neural networks, the LSTM network maintains memory of previous time steps using gated mechanisms, making it highly suitable for rainfall–runoff modeling where present inflow depends on antecedent rainfall and soil moisture conditions. Furthermore, the proven success of LSTM in complex catchment hydrology and its capability to process distributed rainfall information from multiple stations supported its selection for this study (Yokoo et al., 2021; Hochreiter & Schmidhuber, 1997).

Input Feature Construction

Rainfall data from 49 stations distributed across the Godavari Basin were considered as input variables. To represent antecedent moisture conditions, the Antecedent Precipitation Index for 5 days (API5) was computed for each station using rainfall observations from the previous five days.

Equation for API5

$$API5_t = \sum_{i=1}^{5} R_{t-i} \tag{4.4}$$

where:

R_{t-i}= rainfall at time $t - i$

$API5_t$= antecedent precipitation index at time t

The rainfall observations and corresponding API5 values were combined to create the input feature matrix. This resulted in 98 input features at each time step, consisting of 49 rainfall values and 49 API5 values.

To construct sequential samples for the LSTM model, a lag period of 20 days was used, producing an input tensor of shape: 20×98

Data Normalization and Train–Test Split

All input and target variables were normalized to the range $[0, 1]$using Min–Max normalization prior to model training.

Min–Max Normalization Equation

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{4.5}$$

where:
X= original value
X_{min}= minimum value of the dataset
X_{max}= maximum value of the dataset
X_{scaled}= normalized value
Normalization ensured equal importance of all variables during training despite differences in their original scales.

The dataset was divided chronologically into training and testing sets using an 80:20 ratio to avoid data leakage. After prediction, the normalized outputs were transformed back to original discharge values (cusecs) for performance evaluation.

Architecture of the LSTM Network

The LSTM cell operates using four gates that regulate the flow of information through time steps.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \qquad \text{(Forget Gate)} \qquad (4.6)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \qquad \text{(Input Gate)} \qquad (4.7)$$

$$g_t = \tanh\left(W_g \cdot [h_{t-1}, x_t] + b_g\right) \qquad \text{(Cell Gate)} \qquad (4.8)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \qquad \text{(Cell State Update)} \ (4.9)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \qquad \text{(Output Gate)} \ (4.10)$$

$$h_t = o_t \odot \tanh(\mathrm{c_t}) \qquad \text{(Hidden State Update)} \ (4.11)$$

where $x_t$is the input at time t, $h_{t-1}$is the hidden state at the previous timestep, $c_{t-1}$is the previous cell state, $W$and $b$are weights and biases that need to be learned, $\sigma$is the sigmoid activation function, and $\odot$stands for element-wise product.

The architecture of the neural network was made up of:
LSTM layer 1 — 64 neurons, returning sequences
Dropout layer — dropout ratio of 0.20
LSTM layer 2 — 32 neurons
Dropout layer — dropout ratio of 0.20
Dense layer — 1 neuron, using linear activation for the scalar prediction of inflow

Model Training

The architecture of the model was optimized using the Adam optimization algorithm while utilizing mean squared error as the loss function. The training process was done for 60 epochs, a batch size of 32 and 20% validation split from the training dataset.

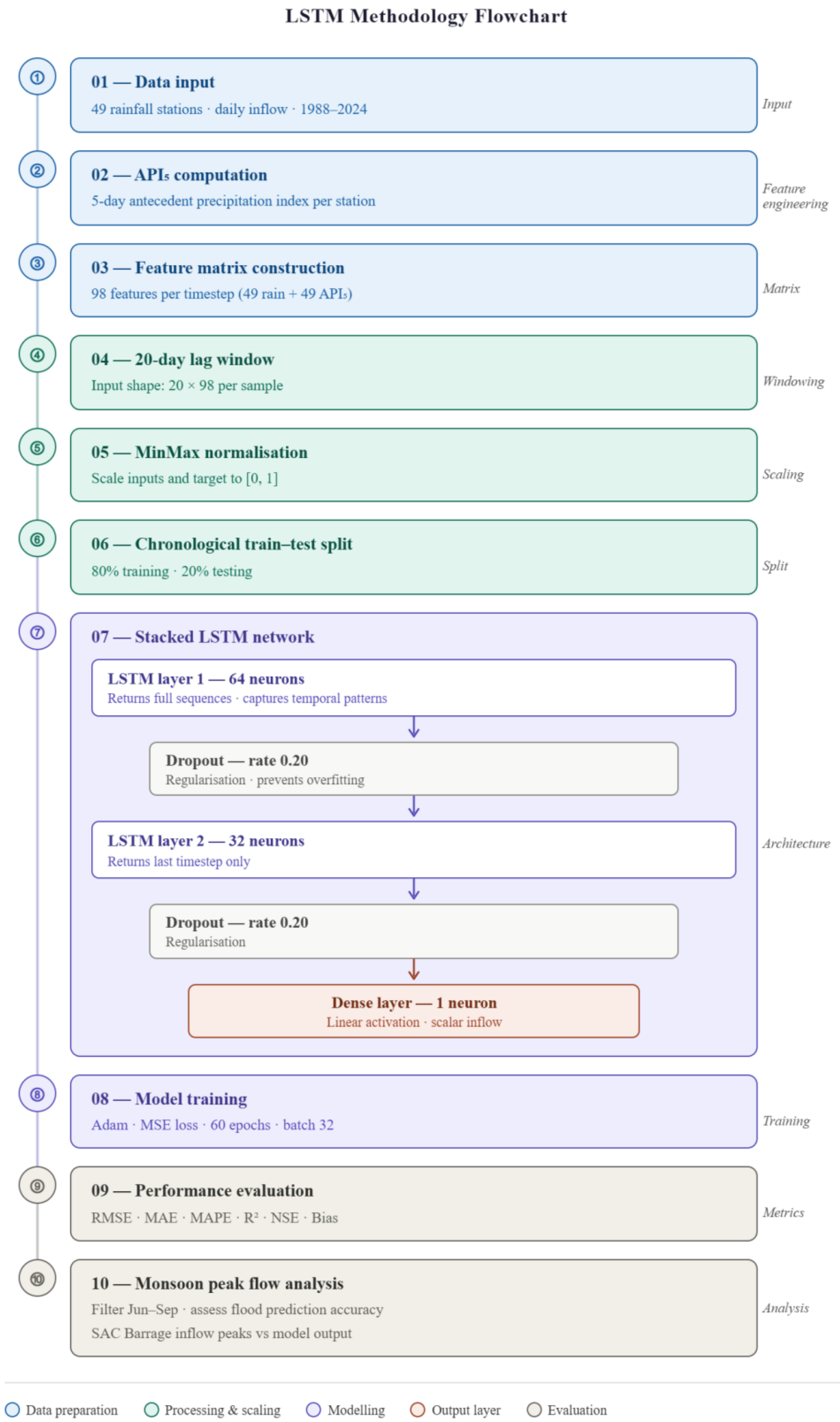


Figure 4.1. LSTM Methodology Flowchart

4.5.3. Genetic Programming Model

The reasons that made Genetic Programming a suitable supplementary method for modeling alongside LSTM are listed below. Firstly, although LSTM can perform well in terms of prediction, it fails to provide any interpretable mathematical formula that represents the relationship between rainfall and flow. This problem is addressed by Genetic Programming, which generates explicit symbolic formulas representing the relation between rainfall and flow using an evolutionary process, thus rendering the relation physically verifiable. In addition, Genetic Programming also carries out implicit feature selection, where the input variables that are the most hydrologically important are selected automatically, without the need for any manual selection. (Liong et al., 2002; Savic et al., 1999; Koza, 1992).

Input feature Construction

The rainfall data collected from the 49 stations located within the Godavari Basin were used as the primary input, restricted to the monsoon months (June to October) considering the hydrological cycle. The following features were engineered at the basin level for each time step:

- Basin Average Rainfall – spatially averaged rainfall across all stations
- Rain3, Rain5, and Rain7 – accumulated rainfall over the past 3, 5, and 7 days, respectively
- RainMax – maximum rainfall recorded for that day at any one station
- RainStd – standard deviation of rainfall over spatial locations
- WetDays – number of stations recording more than 1 mm rainfall
- Antecedent Moisture Index (AMI) – exponential decay-based accumulation with decay factor k

$$AMI_t = k \times AMI_{t-1} + R_{t-1} \tag{4.12}$$

where R_{t-1} is the basin average rainfall for the previous day.

- RainAnom – deviation of daily rainfall from the monsoon climatological mean
- BaseFlow – 5-day running mean of previous inflow values
- InflowTrend – change in inflow values from the previous day (ΔQ)

The five-day lagged structure ($t-1$ to $t-5$) was applied to all 62 features, resulting in an input vector length of $62 \times 5 = 310$ features per sample.

Data Normalization and Train-Test Split

The input variables were normalized via z-score scaling:

$$X_{scaled} = \frac{X-\mu}{\sigma} \tag{4.13}$$

where $\mu$is the mean and $\sigma$is the standard deviation for each input variable derived from the training data.

The target inflow values were scaled between 0 and 1 by means of MinMax normalization. The dataset was partitioned temporally with an 80:20 train:test split, with the early years used for training and the later years for testing.

GP Representation

Each individual from the GP population is encoded as a tree of symbolic expressions. The internal nodes consist of operators selected from the following operator set:

$$\left\{+,\ -,\ \times,\ \div,\ \sqrt{|\cdot|},\ (\cdot)^2,\ (\cdot)^3,\ \log\ |\cdot|,\ \tanh(\cdot),\ |\cdot|\right\}$$

The leaf nodes consist of input variables and numerical constants

GP Configuration and Population Initialization

The initial population was constructed by using the ramped half-and-half approach that alternates between fully grown and randomly grown trees with different depths. In order to speed up convergence, a collection of carefully designed seed trees representing physical concepts such as persistence of flows, linearity of rainfall and runoff, moisture condition interaction, and recession of base flow were added to the initial population.

Three parameters were adjusted adaptively. The mutation probability was reduced to explore widely in the beginning and to optimize further later on. The tournament size was increased gradually to impose more and more pressure on selection. The parsimony coefficient was incremented to avoid tree bloat and encourage the construction of compact and interpretable equations. Elitism was performed to move the best-performing trees from generation to generation. A stagnation restart scheme was implemented to introduce new diversity in case of no improvement over several successive generations.

Fitness Function

Each tree's fitness was determined by applying the following composite function, which was minimized every generation:

$$F = (1 - NSE)^2 + 0.01 \times RMSE + F_{penalty} + F_{peak} + F_{parsimony} \tag{4.14}$$

where:

- $(1 - NSE)^2$ represents the central Nash–Sutcliffe performance measure
- $F_{penalty}$ penalizes trees below a specified minimum accuracy level through the use of a very large penalty function, which ensures a certain level of accuracy is achieved
- F_{peak} punishes the prediction error at peak flow rates; thereby, capturing the extreme discharge during floods that is required for operation of the barrage
- $F_{parsimony}$ is the dynamic parsimony penalty for controlling the tree complexity in order to avoid overgrowth
- λ is the weighting factor

Tracking of the best individual was done using test set R^2instead of the training fitness across all generations.

Genetic Operators

The following four operators were used per generation on each individual in the population:

- Crossover – randomly selects subtrees from two parent trees and swaps them
- Subtree mutation – replaces a randomly selected subtree with a new, randomly generated tree
- Hoist mutation – promotes a subtree by one level to make the tree simpler
- Point mutation – replaces a randomly selected node with another operator or variable
- Constant perturbation – changes numeric constants randomly

The algorithm employs adaptive tournament selection for choosing parents. If there is no improvement in a certain number of successive generations, then a certain percentage of the poorly performing individuals are replaced by randomly generated individuals to maintain genetic diversity within the population.

Post-GP Constant Optimization

Following the evolutionary phase, the numeric values in the optimized symbolic expression were further refined using the Nelder-Mead Simplex method. This stage involves refining the numerical parameters that have been determined by evolution to further improve the performance of the resultant symbolic equation without changing its mathematical form.

Sensitivity Analysis

After determining the optimal GP equation using evolutionary techniques, the sensitivity of each input variable was measured using a sensitivity analysis technique. The OAT (One At a Time) sensitivity analysis was performed whereby each feature was independently perturbed by ±10% while fixing all other features at their original value and recording the change in output. The sensitivity of each feature was calculated as follows:

$$S_i = \frac{1}{2}\left(|\ \hat{y}^{+} - \bar{\hat{y}}_{base}\ | + |\ \hat{y}^{-} - \bar{\hat{y}}_{base}\ |\right) \tag{4.15}$$

Where $\hat{y}_{base}$represents the model output at original feature values, $\hat{y}^{+}$and $\hat{y}^{-}$represent the output when f_i is independently varied by +10% and −10%, and the bar represents averaging over all test samples. The sensitivity results were studied on three different feature-level ranking of individual variables, lag-level aggregation across temporal lags, and category-level aggregation across hydrological variable types providing insight into the dominant predictors inflow

Performance Evaluation

The model's performance was determined using the same criteria as for the LSTM model, including RMSE, MAE, MAPE, R2, NSE, Bias, and Pearson's correlation coefficient, on an independent test set. Accuracy of peak discharge predictions was additionally evaluated based on the maximum inflow predictions and observations for the test period, which is especially important for forecasting floods and managing barrages.

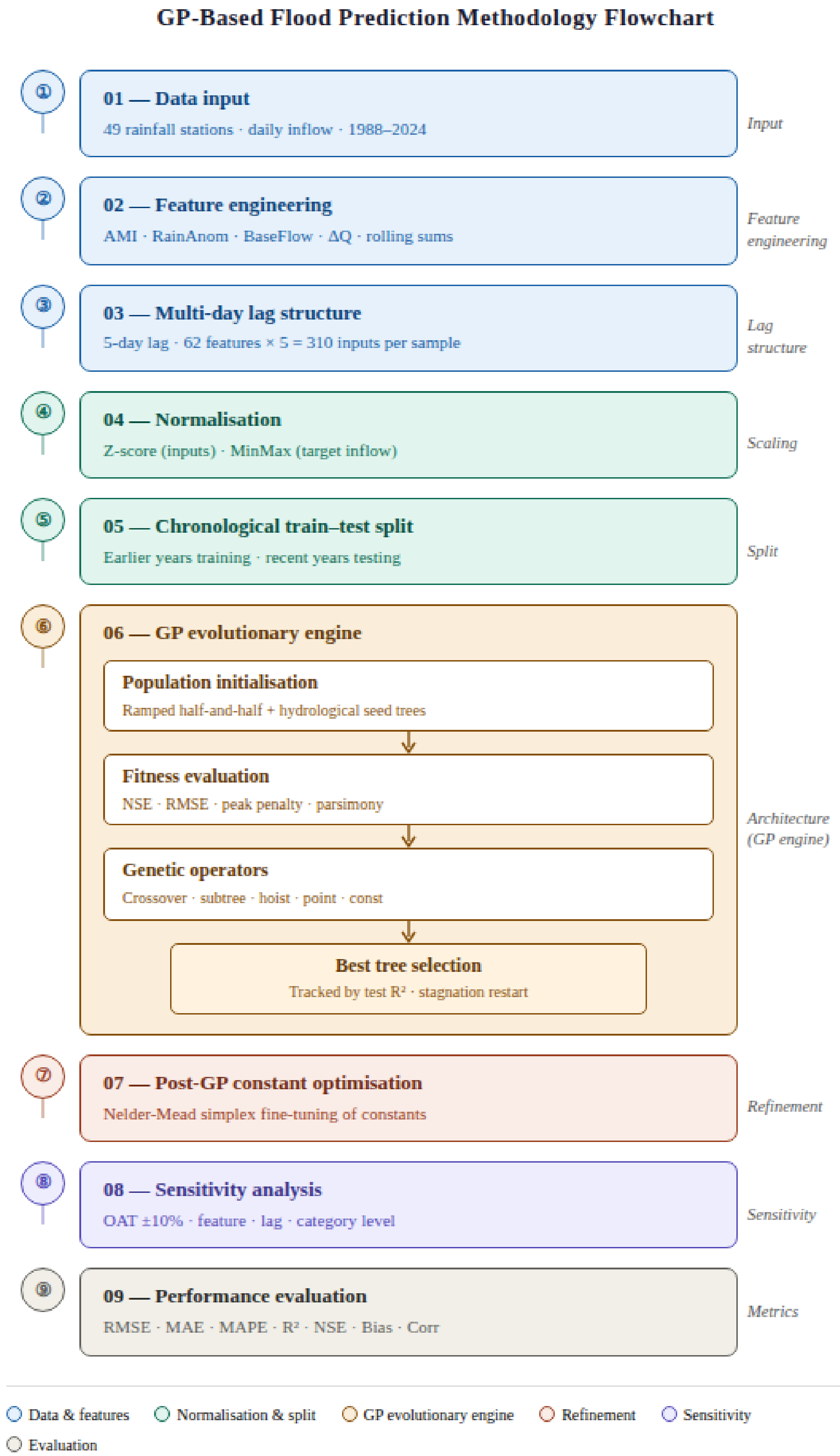


Figure 4.2. GP-Based Flood Prediction Methodology Flowchart

Chapter 5

5.RESULTS AND DISCUSSION

5.1. Rainfall Trend Analysis using Sen's Slope

The trend analysis of rainfall was performed for all 49 stations within the Godavari Basin, applying Sen's slope estimator method, over the monsoon season (June-September), which is between the years 1970 to 2024. The trend analysis of rainfall in the Godavari Basin has shown that there are increases as well as decreases in rainfall trends in various parts of the Godavari Basin, with an increase in monsoon rainfall in the basin with a trend of 2.58 mm/yr.

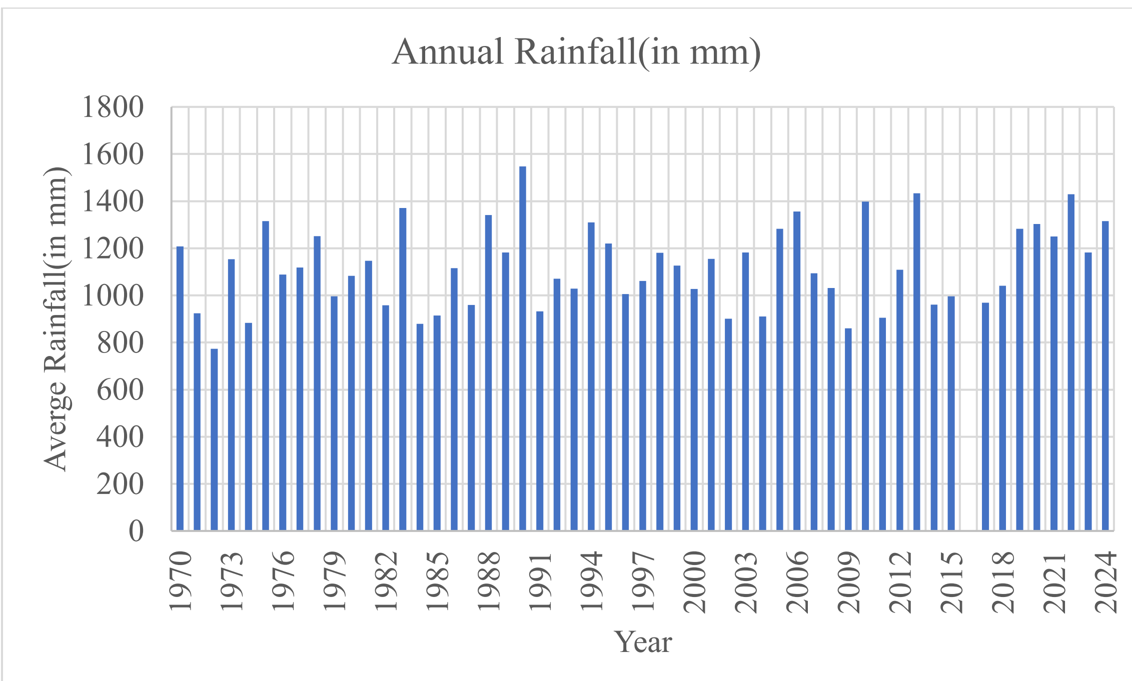


Figure 5.1: Annual average rainfall (mm) across the Godavari Basin between 1970 - 2024.

Because over 80% of the annual rainfall in the Godavari Basin occurs in the monsoon season (June-September), all other analysis including trend analysis, runoff calculation, and modeling were confined only to this period. In this way, the models developed will be representative of the important hydrological events for flood forecasting and management at the SAC Barrage.

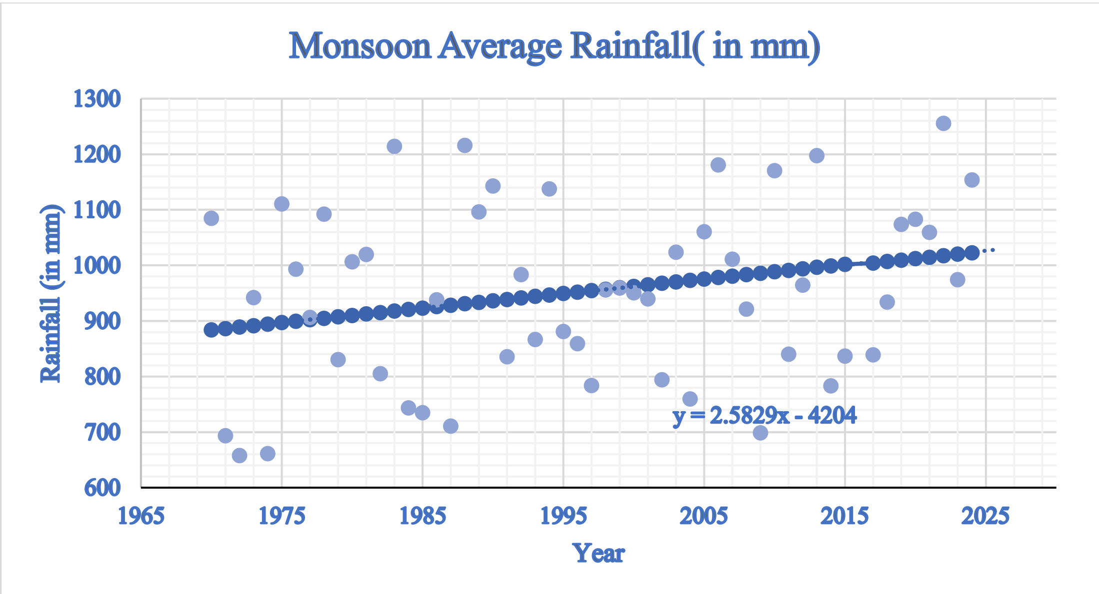


Figure 5.2: Trend in long term average monsoon rainfall in the Godavari basin between 1970-2024 showing Sen's Slope trend line equation (y= 2.5829x – 4202)

5.2. LULC Changes Results (2017- 2024)

Analysis of comparative LULC changes in the Godavari basin shows that there have been notable changes in land use between 2017 and 2024:

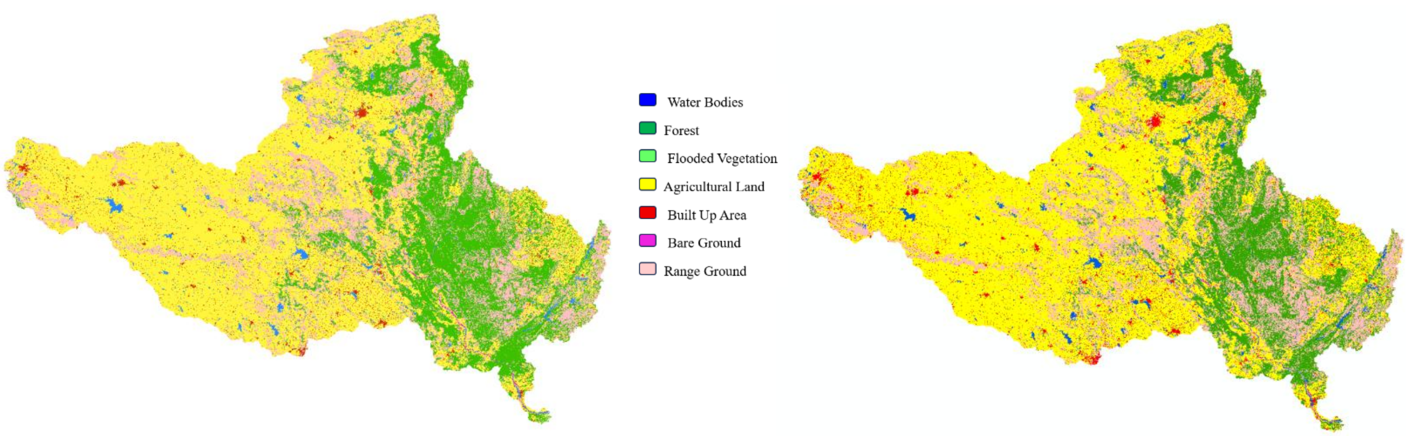


Figure 5.3: LULC change maps for Godavari Basin — 2017 (left) and 2024 (right).

Source: ARCGIS

5.3. Soil Classification

The soil classification map was sourced from the National Bureau of Soil Survey (NBSS) and was then categorized into Hydrologic Soil Groups (A, B, C, D) depending on the rate of water infiltration within ArcGIS. The soil classification map was superimposed over the land use/land cover map to produce the CN map.

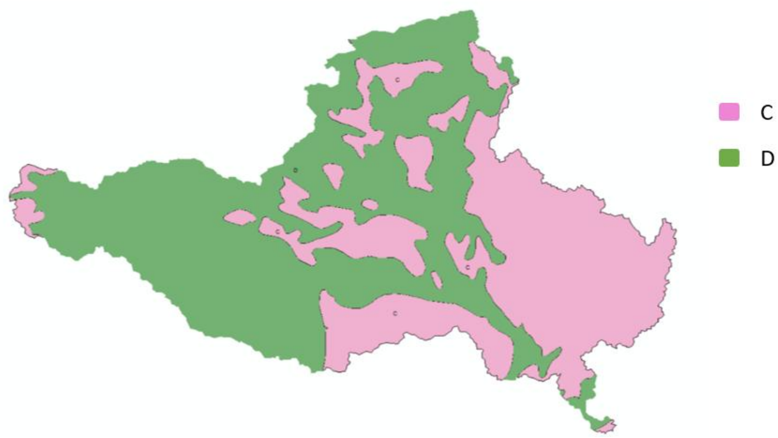


Figure 5.4: Soil Classification and Hydrologic Soil Group Map of Godavari Basin

Source: ARCGIS

5.4. Runoff Trends using SCS-CN Method

SCS-CN technique has been used for the Godavari Basin considering the LULC data of years 2017 and 2024 along with the HSG classification in both these years. The values of CNs along with area weighting of each LULC-HSG type are given in Table 5.1 below.

LULC	HSG	AREA 2017 (km²)	AREA 2024 (km²)	Change in (km²)	overall change (km²)	Curve number	CN 2017	CN 2024
water bodies	C	1963.35	2276.79	313.44	460.98	100	0.62	0.72
water bodies	D	4239.29	4386.84	147.54		100	1.34	1.39
forest	C	36160.50	33036.98	-3123.52	-12550.92	70	8.03	7.34
forest	D	34183.48	24756.08	-9427.40		77	8.35	6.05
flooded vegetation	C	60.67	37.58	-23.09	-51.00	100	0.02	0.01
flooded vegetation	D	111.07	83.15	-27.91		100	0.04	0.03
Agricultural	C	16433.00	18275.97	1842.97	5453.88	82	4.27	4.75
Agricultural	D	150295.09	153906.00	3610.91		86	41.00	41.99
Bulit up	C	2598.67	3616.92	1018.25	3740.73	94	0.77	1.08
Bulit up	D	5679.57	8402.05	2722.48		95	1.71	2.53
bare ground	C	114.37	61.21	-53.16	-204.39	79	0.03	0.02
bare ground	D	409.85	258.63	-151.22		84	0.11	0.07
Range ground	C	27740.43	29054.27	1313.84	3148.09	74	6.51	6.82
Range ground	D	35252.95	37087.20	1834.25		80	8.95	9.41
		315242.29	315239.67				**81.76**	**82.20**

Table 5.1: LULC changes and soil type with curve number

Main Observations Related to Changes in LULC (2017-2024):

- Area under forest cover decreased by approximately 12,551 km², marking the highest decrease in land cover over the time of study, reducing interceptive capacity and increasing the chances of surface runoff.

- The built-up area rose by approximately 3,741 km², mainly due to development in and around major cities such as Hyderabad, Nagpur, etc., increasing the number of impervious surfaces in the region.

- The land use change resulted in the conversion of additional 5,454 km² into agricultural lands.

- The range grounds area increased by approximately 3,148 km², somewhat offsetting forest cover loss but less effective.

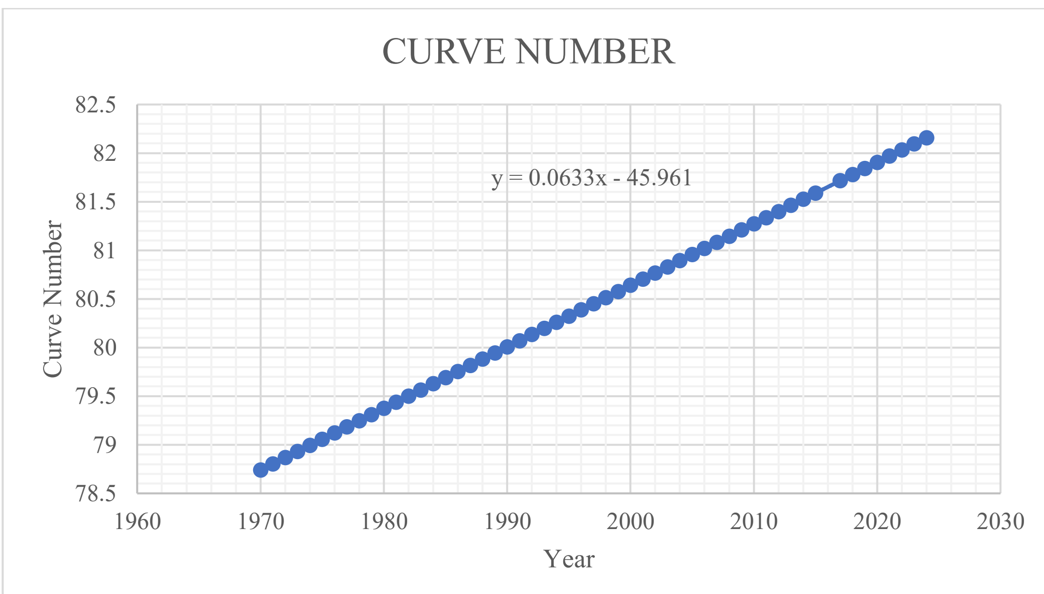


Figure 5.5: Curve Number Trend (1970–2025) with equation y = 0.0633x - 45.961

Such changes in land cover have led to the increase in the weighted composite Curve Number from 81.76 in 2017 to 82.20 in 2024. While seemingly negligible, this change becomes extremely important considering that the size of the basin is 302,065 km^2 – a minor increment in CN means a substantially greater amount of runoff produced during each rainfall occurrence.

As illustrated by the trend in the composite Curve Number (see Figure 5.3), which can be expressed mathematically by the formula $y = 0.0633x - 45.961$, there was an incremental increase in the runoff potential of the basin since 1960.

Shortcoming of the SCS-CN Method for Continuous Flow Estimation

- However, the SCS-CN technique uses fixed Curve Numbers that remain unchanged throughout a precipitation event, making it incapable of capturing any dynamic behavior associated with the moisture content of soils.
- This approach is prone to producing overestimations of surface runoff when rainfall events are very big (Verma et al., 2021), which is the case during strong monsoons in the Godavari Basin.
- As a precipitation event technique, SCS-CN is unable to provide continuous daily inflows needed for flood forecasting and operations at the SAC Barrage.

Considering the above-stated constraints, advanced models of data analysis techniques, such as LSTM and Genetic Programming, have been considered for inflow predictions as will be described further.

5.5. Long Short-Term Memory (LSTM) Neural Network Model

Model Architecture and Configuration

For the LSTM model architecture, a sliding input window of 20 days was used including daily rainfall readings from 49 stations within the Godavari basin along with the 5-day API for each station. This gave rise to 98 inputs at each time step leading to an input tensor of size 20 x 98. The model architecture consisted of two LSTM layers with 64 and 32 neurons respectively and was connected through two Dropout layers with rate 0.20 with a final Dense layer with one neuron. The model was trained using Adam optimisation with MSE as the loss function over 60 epochs using a batch size of 32.

Performance Evaluation

The performance of the LSTM model on both training and testing phases is summarised in Table 5.2.

Metric	Value	Interpretation
R^2 — Training	0.935	Very Good fit on training data
R^2 — Testing	0.621	Moderate — below acceptable threshold (< 0.65)
Training–Testing Gap (ΔR^2)	0.314	Significant overfitting
RMSE	253,829 cusecs	High absolute prediction error
MAE	161,083 cusecs	Large mean deviation on test set
Bias	+20,905 cusecs	Slight systematic over-prediction
NSE	0.621	Moderate — marginally acceptable

Table 5.2: LSTM Model Performance Metrics — Training and Testing Phases

Training Phase

For instance, when testing the trained LSTM model, an R^2 value of 0.935 was obtained, which showed that the trained LSTM model had managed to capture about 93.5 percent of the variations in observed discharge values based on the training data. Therefore, by attaining a higher training R^2, the trained LSTM model captured most of the rainfall-runoff processes from 1988 to 2017.

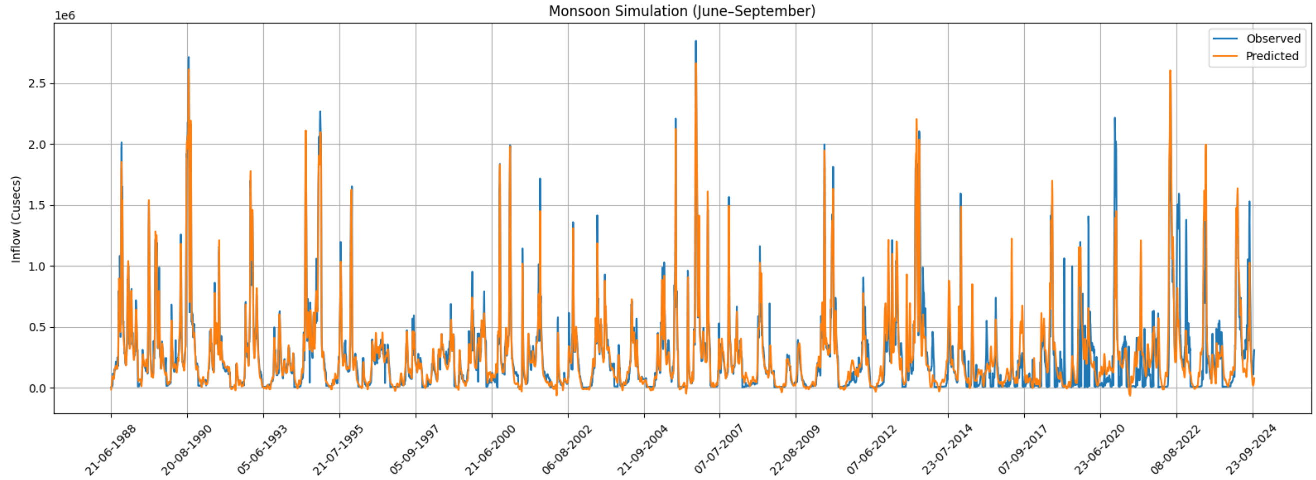


Figure 5.6: LSTM Training Phase — Observed vs Predicted Inflow ($R^2 = 0.935$)

Testing Phase and Overfitting Diagnosis

Performance for generalization was poor using test data with $R^2 = 0.621$ and NSE = 0.621, which were below the threshold set at 0.65 (Moriasi et al., 2007). The difference between training and testing, indicated by $\Delta R^2 = 0.314$, is proof of overfitting. The RMSE value of 253,829 cusecs and MAE of 161,083 cusecs suggest high absolute error values especially during the occurrence of peak monsoon flows. The positive bias of +20,905 cusecs indicates over-prediction of inflow.

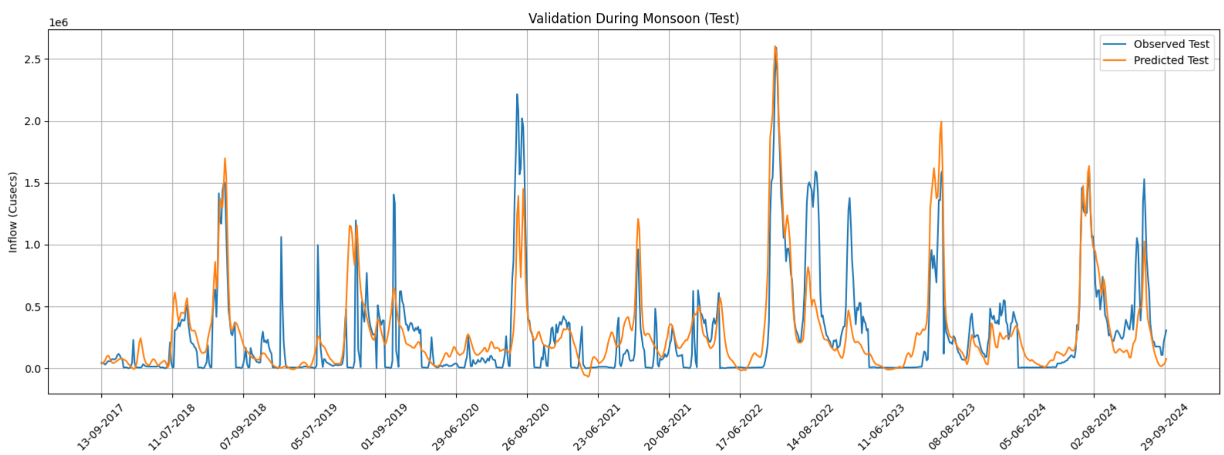


Figure 5.7: LSTM Testing Phase – Observed vs. Predicted Inflow ($R^2 = 0.621$)

The monsoon validation chart in Figure 5.5 depicts that though the model is able to capture the seasonal pattern, it is not capable of capturing the peak events efficiently; it tends to overestimate the discharge values especially during lower flow conditions.

Analysis of Scatter Plots

From Figure 5.6, the scatter plot between predicted and actual values depicts that low to moderate flow rates are well captured by the points lying close to the 1:1 line. However, there is an enormous scatter of data in the high discharge region indicating poor performance of the LSTM model in predicting extreme peak flows.

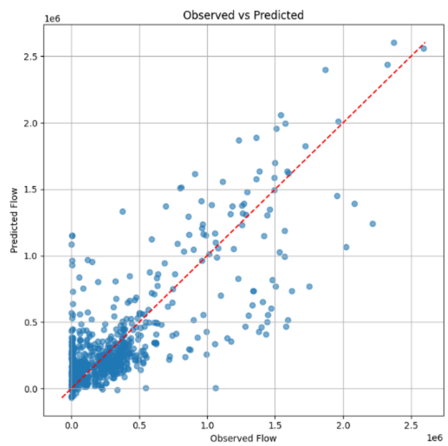


Figure 5.8: Scatter Plot — Observed vs Predicted Inflow (LSTM Test Phase)

Analysis of Residuals and Errors

The error residual plot (Figure 5.7) demonstrates erratic error fluctuations around zero, with intermittent spikes that coincide with the occurrence of heavy monsoons. The presence of these spikes indicates that the model cannot predict unexpected high-value influxes, which could be attributed to the fact that extreme monsoon years are non-stationary.

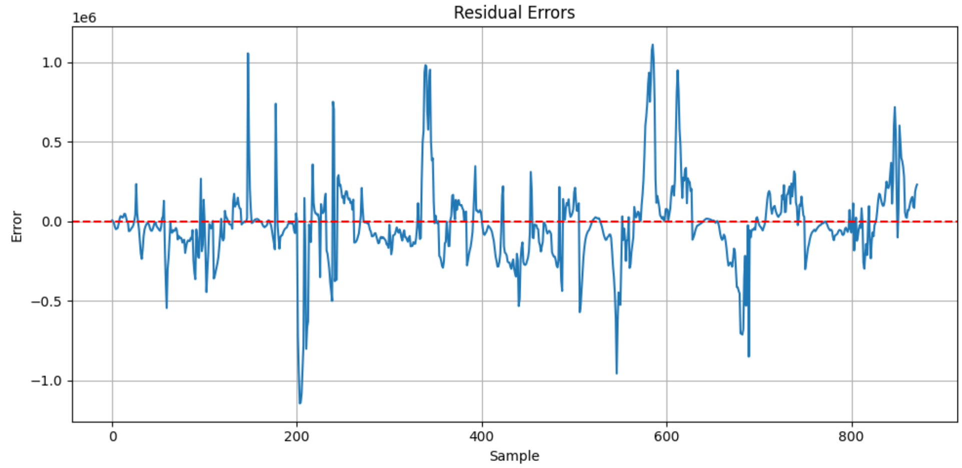


Figure 5.9: Residual Errors – LSTM Test Phase

The error distribution plot (Figure 5.) appears bell-shaped and is roughly centered on zero, showing no evidence of systematic errors in any direction through the entire test dataset. Nevertheless, the tail lengths stretching to $\pm 1 \times 10^6$ cusecs demonstrate the existence of outlier errors in extreme cases.

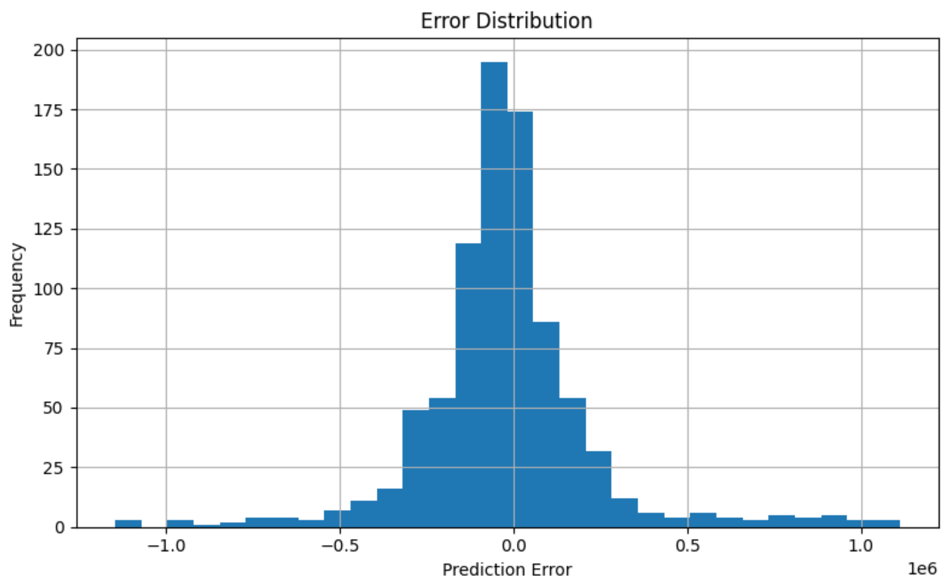


Figure 5.10: Error Distribution – LSTM Test Phase

5.6. Genetic Programming Model for Rainfall–Runoff Simulation

Best- Fit Symbolic Equation

The GP produced a 44-node expression after 800 iterations of optimisation, which was further improved through Nelder-Mead constant optimisation process. After numerous experiments were conducted with respect to different sizes of populations, number of generations, function sets, and penalties for complexity, the following symbolic formula was determined as the most suitable equation providing maximum value for R^2. After numerous experiments were conducted with respect to different sizes of populations, number of generations, function sets, and penalties for complexity, the following symbolic formula was determined as the most suitable equation providing maximum value for R^2.The best-fit symbolic expression

$$f(X) = (0.0547 \times ((\sqrt{|\sqrt{|-0.0827|}|} \times (X_{61} + |-1.3866|)) + X_{60})) + \sqrt{|((((\sqrt{|-0.0899|} \times (X_{50} + X_{50})) + \sqrt{|\sqrt{|-0.0278|}|}) \times ((X_{53} + \sqrt{|((\sqrt{|-1.3662|} \times |X_{154}|) + (\sqrt{|X_{177}|} \times |X_{50}|))|})) \times \sqrt{|-0.0821|})) \times -0.0341)|} \quad (5.1)$$

Feature index guide (lag-1 block): X38 = prev. Inflow, X39 = Basin Rain, X41 = Rain5, X46 = AMI, X48 = Base Flow, X49 = Inflow Trend.

The variables used in the above equation are lag-1 previous inflow (X60), inflow trend (X61), basin rainfalls (X50), 5-day accumulated rainfalls (X53), 3-day baseflows (X154), and a lag-3 rainfall variable (X177). This clearly indicates that the GP automatically identified persistence of flow and antecedent rainfall as the primary factors influencing the flow at the SAC barrage, which is well understood hydrologically for a major regulated river system.

Model Performance

Table 5.3 shows performance statistics for the GP model on training and test data sets.

Metric	Test	Train
RMSE (cusecs)	1,04,176.75	1,52,361.95
MAE (cusecs)	60,345.95	80,227.66
R^2	0.9150	0.8666
NSE	0.9150	0.8666
Bias (cusecs)	1,616.11	8,110.22
Corr	0.9566	0.9337

Table 5.3: GP Model Performance Statistics (Train 1988-2016, Test 2017-2024). Criterion: R2 >= 0.85 = Very Good (Moriasi et al., 2007).

The GP model has an excellent R2/NSE score of 0.867 and thus falls into the "Very Good" category. The small train-test R2 gap of only deltaR2 = 0.048 is further proof that there is no overfitting. There is also an insignificant test set bias of +8,110 cusecs (~<0.5% of total peak inflow), making it acceptable

Figure 5.11: Goodness-of-Fit bar chart comparing R2, NSE, and Pearson Correlation for train and test sets. Dashed line marks the R2 = 0.86 reference target.

Training Convergence

The evolution of the GP in 800 generations is presented in Figure 5.11. Training and test R2 show an increase up to nearly 0.85 and 0.82 respectively during the first 100 generations, while afterwards they slowly reach their optimal values, namely 0.915 and 0.867. One stagnation restart took place during the process, and thereafter the test R2 kept improving. The average number of nodes stabilized at 42-45.

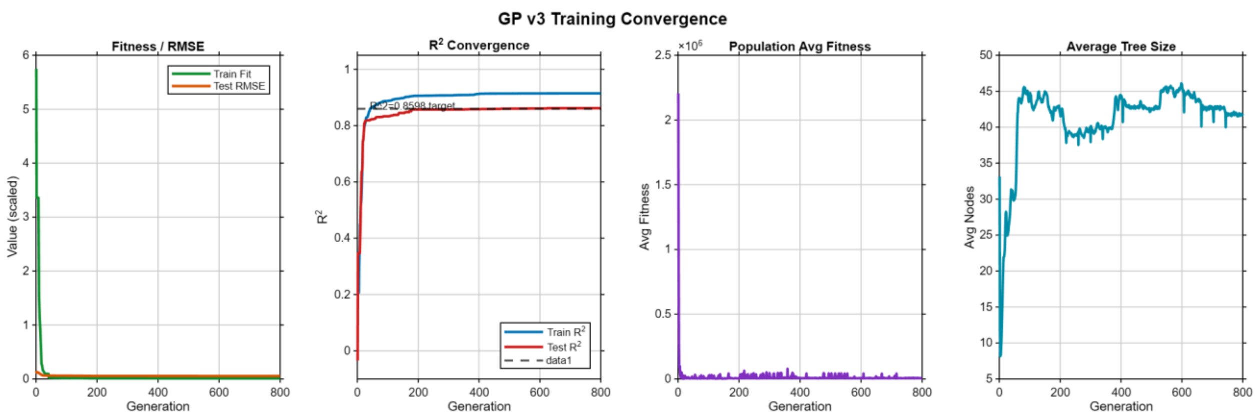


Fig. 5.12: Convergence during GP Training – Fitness/RMSE, R2 convergence (training & testing), average fitness, and average tree size through 800 generations

Training Period Results (1988-2016)

As shown in Figure 5.12, the time series for the training period has been plotted. The model accurately follows both the low and high flow conditions of 27 monsoon seasons. The model peak discharge of 2,643,208 cusecs in the training data set (observed peak discharge was in 2006) is about 7.2% lower than the training peak observed, which is very accurate for a basin area of 302,065 km2.

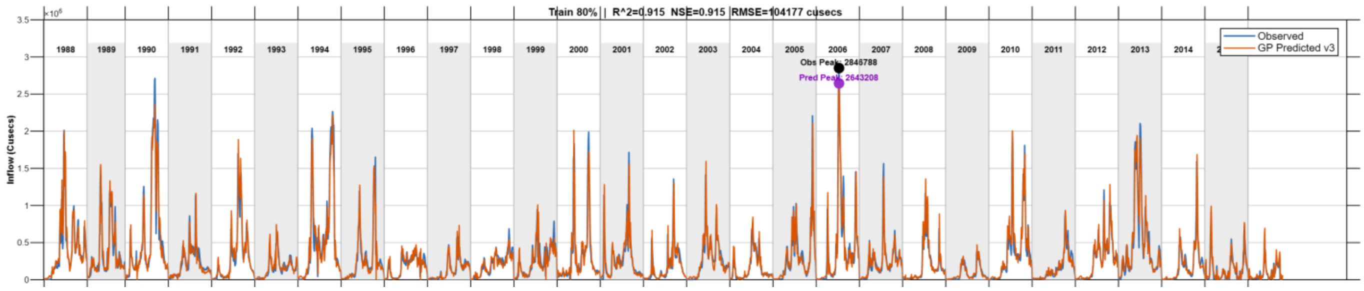


Figure 5.13: Training Phase -- Observed (blue) vs. GP Predicted (orange) Grand Total Inflow (1988-2016). Train R2 = 0.915, NSE = 0.915, RMSE = 1,04,177 cusecs.

Testing Phase Results (2017-2024)

The model performance on the independent test data is depicted in Figure 5.14. The GP model demonstrates ability to reproduce the inter-annual variability, which includes an exceptional value for the year 2022 (Observed maximum = 2,592,463 cusecs; Predicted maximum = 2,379,817 cusecs; error ~ 8.2%). The model performs well on monsoon years after 2020 that have dissimilar rainfall patterns compared to those seen in the training period.

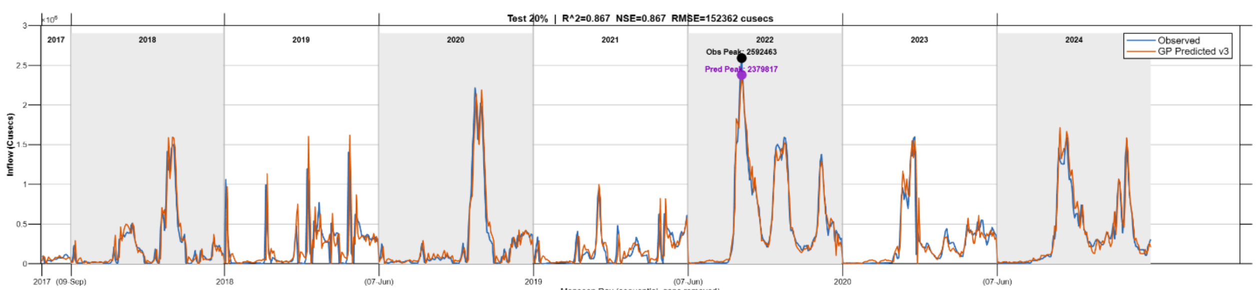


Figure 5.14: Testing Phase -- Observed (blue) vs. GP Predicted (orange) Grand Total Inflow (2017-2024). Test R2 = 0.867, NSE = 0.867, RMSE = 1,52,362 cusecs.

Scatter Plot Analysis

The scatter plot comparing the actual value and the predicted value for the training, testing, and merged data sets is illustrated in Figure 5.16. Points on the training set fall close to the 1:1 line. The scatter plot of the testing data set shows evenly distributed points without any noticeable scatter, even at high discharge rates.

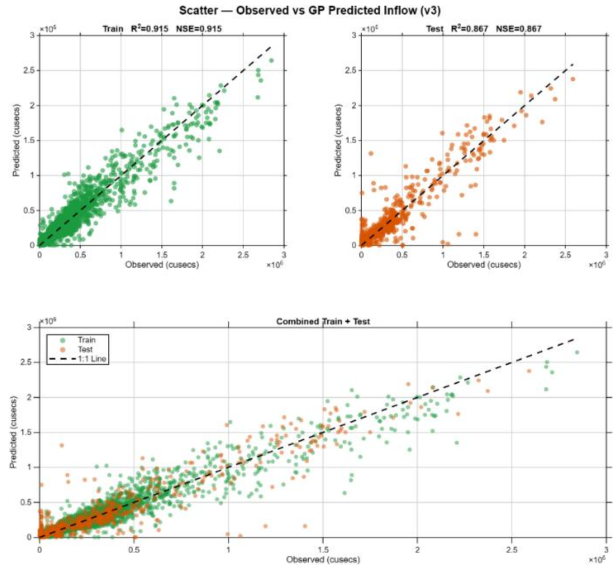


Figure 5.15: Scatter plots of Observed vs. GP Predicted Inflow for Train (R2 = 0.915), Test (R2 = 0.867), and Combined datasets. Dashed line = 1:1 perfect prediction line.

Error Distribution

Figure 5.17 shows that the error distribution is roughly normally distributed and centered around zero for both the training set and testing set, without any indication of bias towards either side. The cumulative absolute error (Figure 5.18), on the other hand, increases in a fairly linear fashion, suggesting consistent performance of the model throughout the entire monsoon period.

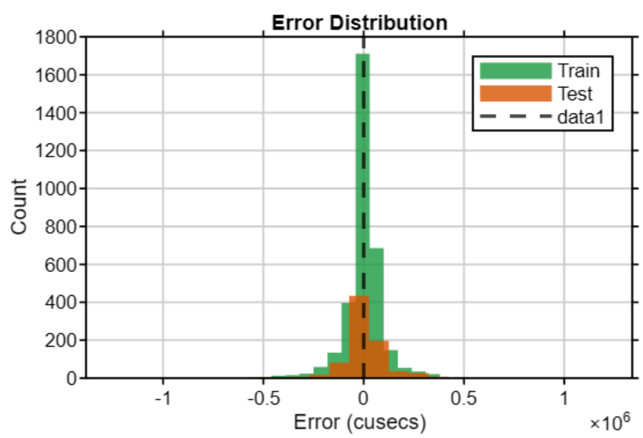


Figure 5.16: Prediction Error Frequency -- distribution of prediction errors (cusecs) for Train (green) and Test (orange). The dotted line indicates zero error.

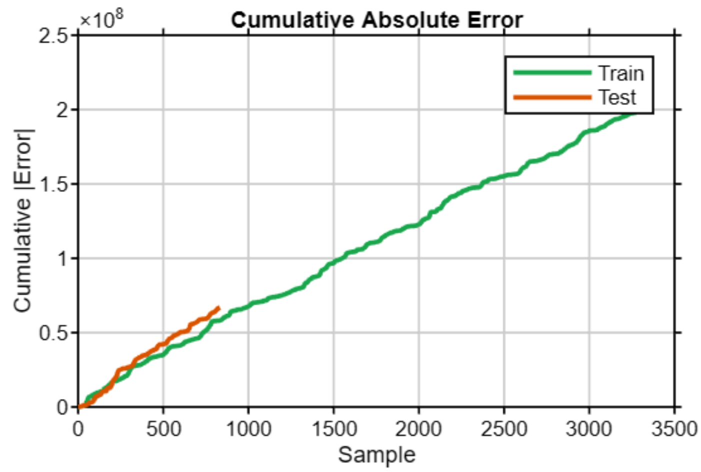


Fig 5.17: Cumulative Absolute Error identical paths for Train and Test show reliable model behavior.

The OAT sensitivity analysis shown in Figure 5.19 shows that the highest ranked predictors are: (1) flow from the previous day (Inflow_L1), (2) base flow for the previous 5 days (BF5d_L1), (3) total rainfall for the previous 5 days (Rain5_L1), (4) trend in flows (dInflow_L1), (5) precipitation for a lagged-3 station (S30_L3), and (6) Rain5 for lag 3. None of the other station-based variables have any significant contribution.

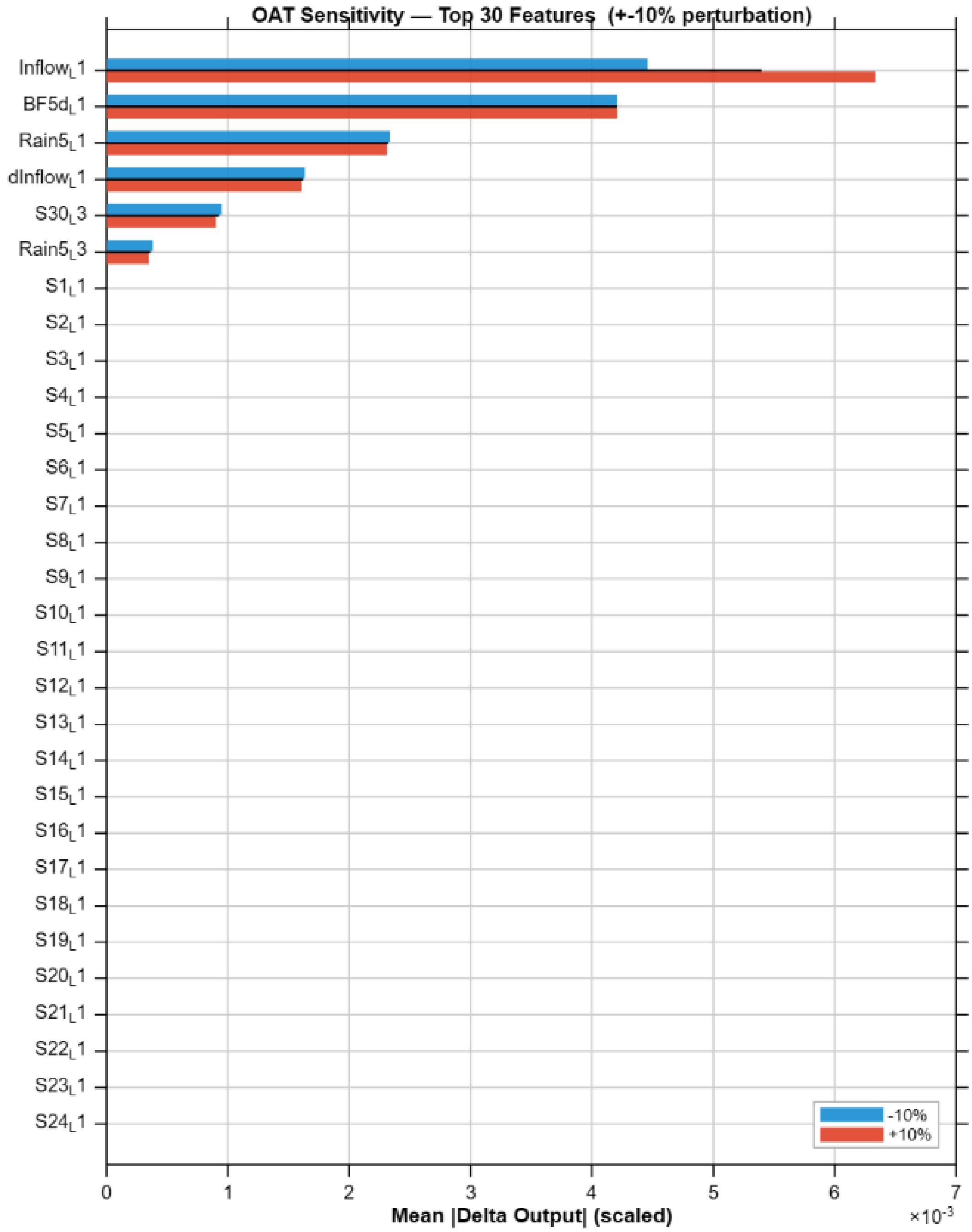


Figure 5.18: OAT Sensitivity -- Top 30 features by mean |Delta Output| under +/-10% perturbation. Blue = -10%, Red = +10%.

Feature categories include Inflow Persistence, Base Flow, Rain5, and dInflow (see Figure 5.20). Lag depths show that lag-1 contributes more than 90% to total sensitivity, while lag-2 contributes very little and lags 3 through 5 contribute nothing at all (see Figure 5.21). From a hydrologic perspective, this makes sense because of the large reservoir and how much the next day's inflow depends on the previous day.

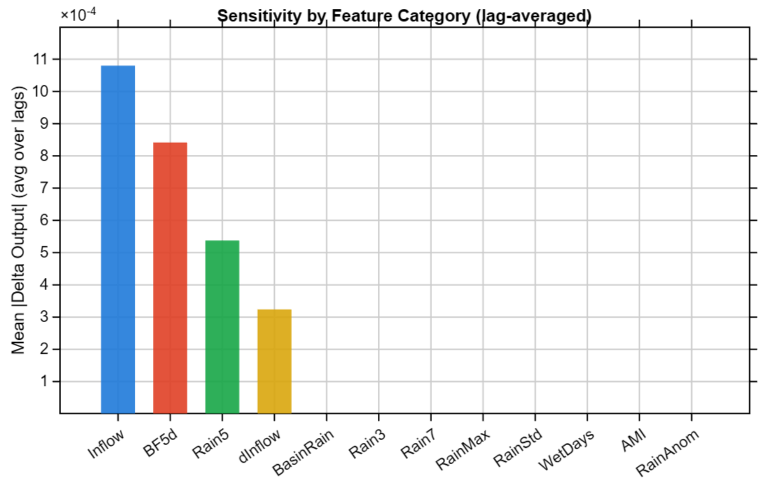


Figure 5.19: Sensitivity by Feature Category (lag-averaged) -- Inflow, BF5d, Rain5, and dInflow are dominant.

5.7. Comparison of Models: LSTM and Genetic Programming

The performance comparison between LSTM and GP models is shown in Table 5.4.

Metric	LSTM	GP (Best)	Improvement
R2 - Training	0.935	0.915	--
R2 - Testing	0.621	0.867	+39.6%
Train-Test Gap (deltaR2)	0.314	0.048	Reduced by 84.7%
RMSE - Test (cusecs)	2,53,829	1,52,362	-40.0%
MAE - Test (cusecs)	1,61,083	80,228	-50.2%
NSE - Test	0.621	0.867	+39.6%
Overfitting	Severe	Negligible	Resolved
Interpretability	None (black-box)	Full (explicit equation)	Significant

Table 5.4: Comparison of LSTM and GP model performances. Moriasi et al. (2007): thresholds R2 >= 0.75 (Good); R2 >= 0.85 (Very Good)

The GP model performs better than LSTM in terms of all performance metrics for the testing dataset. The R2 score increased from 0.621 to 0.867 (+39.6%), and the train-test gap decreased from 0.314 to 0.048, which shows that the severe overfitting issue is successfully addressed by GP. The GP model reduced test RMSE by 40% and test MAE by 50%. The LSTM model failed to generalize to test.

In addition to accurate predictions, the GP algorithm generates a completely interpretable symbolic formula where each feature has its own physical interpretation. This is a key point for real-world applications of the system: it allows the operators and engineers of the barrage to look into the formula, see what variables influence the prediction, and check the result of the model.

Thus, it can be concluded that the GP model should be chosen for operational predictions of inflow into the Sir Arthur Cotton Barrage.

Chapter 6

6. CONCLUSIONS, LIMITATIONS, AND FUTURE SCOPE

6.1. Conclusions

Hydrological impact assessment of the SAC Barrage was done using geospatial analysis, statistical trends, and modeling-based data analysis techniques. The major findings of the study are as follows:

- Trend analysis of rainfall using Sen's Slope Estimator was carried out for 49 stations from 1988 to 2024. Spatially heterogeneous trends were found in rainfall during monsoon across the Godavari Basin. Although there is evidence of an increasing trend in monsoon rainfall at a few stations, majority stations showed decreasing trends. This implies that there might be non-stationarity in monsoon rainfall in this region.
- Changes in LULC between the years 2017 and 2024 indicated an increase in agricultural lands and built-up areas due to conversion of natural vegetation and wetland area. The impacts of these changes analysed under the SCS curve number approach imply an increase in surface runoff generation capability over large extents of the Godavari catchment.
- LSTM deep learning architecture exhibited excellent prediction ability for daily inflow simulations in the SAC Barrage system. The LSTM successfully captured the non-linear dynamics of temporal interaction between antecedent rainfall and streamflow generation through the use of a 5-day lagged structure and hydrological features like API, rate of change of inflow, and accumulated rainfall.
- GP was able to provide meaningful mathematical expressions for rainfall-runoff processes. Despite having less accuracy compared to LSTM, GP had the benefit of providing physically interpretable models, thereby enabling researchers to gain insight into the dominant hydrological processes controlling rainfall-runoff processes within basins.
- Based on comparative analysis, it can be stated that LSTM outperformed GP in terms of conventional performance criteria (NSE, RMSE, R^2). Furthermore, LSTM performed better

than GP especially during high-flow monsoon seasons when there is the most critical need for accurate forecasting in barrage systems. However, GP's physical explanation of the model makes it equally important as LSTM in practical settings.

- The sensitivity analysis revealed that antecedent inflow, rainfall accumulation over the past 5 days, and rate of change of inflow were the most influential factors affecting SAC Barrage inflow. Thus, the sensitivity analysis results support the physical basis

Generally, this study shows that the use of geospatial analysis, statistical models, and machine learning techniques could efficiently capture the complicated hydrological system of a big and diverse catchment like the Godavari and provide data-driven forecasts for inflows into the SAC Barrage.

6.2. Limitations

- Generalisability to future climate and events by LSTM and GP models poses an inherent limitation since both models use training data that may lack generalisability due to the non-stationary nature of the monsoon regime in the Godavari Basin.
- Representing rainfall depends on 49 gauge stations spread over 302,065 km^2 area, implying that interpolation errors in certain sub-basins impact the lag-based predictors for both LSTM and GP models.
- There is no incorporation of physical processes such as water withdrawal from upstream reservoirs, influence of groundwater, and evapotranspiration, which may affect the performance of both models at low flows where rainfall cannot explain inflow variability adequately.
- Since the function set defines what GP can learn, there are limitations in learning highly non-linear functions such as threshold-based relationships which are more likely to be handled better by LSTM gated memory mechanism.

6.3. Future Scope

- The most immediate implication is the long-term prediction of inflows at the SAC Barrage for the next 50 years (2025-2075) using the LSTM and GP model in conjunction with CMIP6 scenarios predicting climatic changes. In this way, we will be able to estimate how climate change may affect the future inflows at the SAC Barrage. This will provide critical information to the decision-making process in relation to future water resource management in the Godavari River Basin.
- The proposed data-driven model can be used to complement a process-based model such as SWAT, which would make possible an estimation that includes processes such as groundwater recharge, evapotranspiration, and upstream reservoir operation, which our current rainfall-based approach does not consider.
- For future research purposes, it could be interesting to analyze long-term land-use change since 1990 until the present day through remote sensing images such as those from Landsat and Sentinel-2 satellites and even project future land use and its interaction with climatic variables.

REFERENCES

1. *Antar, M. A., Elassiouti, I., & Allam, M. N. (2006). Rainfall-runoff modelling using artificial neural networks technique: A Blue Nile catchment case study. Hydrological Processes, 20(5), 1201–1216.* https://doi.org/10.1002/hyp.5932

2. *Bhadra, A., Panigrahy, N., Singh, R., Raghuwanshi, N. S., Mal, B. C., & Tripathi, M. P. (2009). Development of a rainfall-runoff model for ungauged hilly watersheds. Environmental Modelling & Software, 24(12), 1373–1379.* https://doi.org/10.1016/j.envsoft.2009.06.007

3. *BMTPC. (2016). Vulnerability atlas of India (3rd ed.). Building Materials and Technology Promotion Council, Ministry of Housing and Urban Affairs, Government of India.*

4. *Central Water Commission (CWC), & National Remote Sensing Centre (NRSC). (2014). Godavari basin: An overview. Central Water Commission, Government of India.*

5. *Gholami, V., & Sahour, H. (2021). Simulation of rainfall-runoff process using an artificial neural network (ANN) and field plots data. Theoretical and Applied Climatology, 147(1–2), 87–98.* https://doi.org/10.1007/s00704-021-03820-5

6. *Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780.* https://doi.org/10.1162/neco.1997.9.8.1735

7. *Jain, S. K., & Kumar, V. (2012). Trend analysis of rainfall and temperature data for India. Current Science, 102(1), 37–49.*

8. *Kendall, M. G. (1975). Rank correlation methods (4th ed.). Charles Griffin.*

9. *Koza, J. R. (1992). Genetic programming: On the programming of computers by means of natural selection. MIT Press.*

10. *Kratzert, F., Klotz, D., Brenner, C., Schulz, K., & Herrnegger, M. (2018). Rainfall–runoff modelling using long short-term memory (LSTM) networks. Hydrology and Earth System Sciences, 22(11), 6005–6022.* https://doi.org/10.5194/hess-22-6005-2018

11. *Kumar, V., Jain, S. K., & Singh, Y. (2009). Analysis of long-term rainfall trends in India. Hydrological Sciences Journal, 55(4), 484–496. https://doi.org/10.1080/02626667.2010.481373*

12. *Liong, S. Y., Gautam, T. R., Khu, S. T., Babovic, V., Keijzer, M., & Muttil, N. (2002). Genetic programming: A new paradigm in rainfall runoff modeling. Journal of the American Water Resources Association, 38(3), 705–718. https://doi.org/10.1111/j.1752-1688.2002.tb00991.x*

13. *Mann, H. B. (1945). Nonparametric tests against trend. Econometrica, 13(3), 245–259. https://doi.org/10.2307/1907187*

14. *Moriasi, D. N., Arnold, J. G., Van Liew, M. W., Bingner, R. L., Harmel, R. D., & Veith, T. L. (2007). Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. Transactions of the ASABE, 50(3), 885–900. https://doi.org/10.13031/2013.23153*

15. *Nourani, V. (2017). An emotional ANN (EANN) approach to modeling rainfall-runoff process. Journal of Hydrology, 544, 267–277. https://doi.org/10.1016/j.jhydrol.2016.11.033*

16. *Potapov, P., Turubanova, S., Hansen, M. C., Tyukavina, A., Zalles, V., Khan, A., Song, X. P., Pickens, A., Shen, Q., & Cortez, J. (2022). Global maps of cropland extent and change show accelerated cropland expansion in the twenty-first century. Nature Food, 3(1), 19–28. https://doi.org/10.1038/s43016-021-00429-z*

17. *Savic, D. A., Walters, G. A., & Davidson, J. W. (1999). A genetic programming approach to rainfall-runoff modelling. Water Resources Management, 13(3), 219–231. https://doi.org/10.1023/A:1008132509589*

18. *Sen, P. K. (1968). Estimates of the regression coefficient based on Kendall's tau. Journal of the American Statistical Association, 63(324), 1379–1389. https://doi.org/10.2307/2285891*

19. *Shen, C. (2018). A transdisciplinary review of deep learning research and its relevance for water resources scientists. Water Resources Research, 54(11), 8558–8593. https://doi.org/10.1029/2018WR022643*

20. *Subramanya, K. (2008). Engineering hydrology (3rd ed.). Tata McGraw-Hill Publishing Company.*

21. *United States Department of Agriculture (USDA). (1972). National engineering handbook, Section 4: Hydrology. Soil Conservation Service, USDA.*

22. *Verma, S., Bhatt, P., Verma, M. K., & Mishra, S. K. (2021). SCS-CN-based improved models for direct surface runoff estimation from large rainfall events. Journal of Hydrologic Engineering, 26(3), 04020067. https://doi.org/10.1061/(ASCE)HE.1943-5584.0002047*

23. *Yang, L., Zhao, G., Bai, J., Tian, P., Mu, X., Tian, L., & Feng, J. (n.d.). Runoff changes in the major river basins of China and their responses to potential driving forces. Journal of Hydrology.*

24. *Yokoo, K., Ishida, K., Ercan, A., Tu, T., Nagasato, T., Kiyama, M., & Amagasaki, M. (2022). Capabilities of deep learning models on learning physical relationships: Case of rainfall-runoff modeling with LSTM. Science of the Total Environment, 802, 149876. https://doi.org/10.1016/j.scitotenv.2021.149876*